



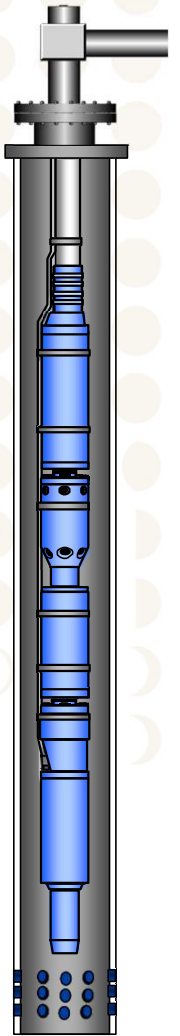
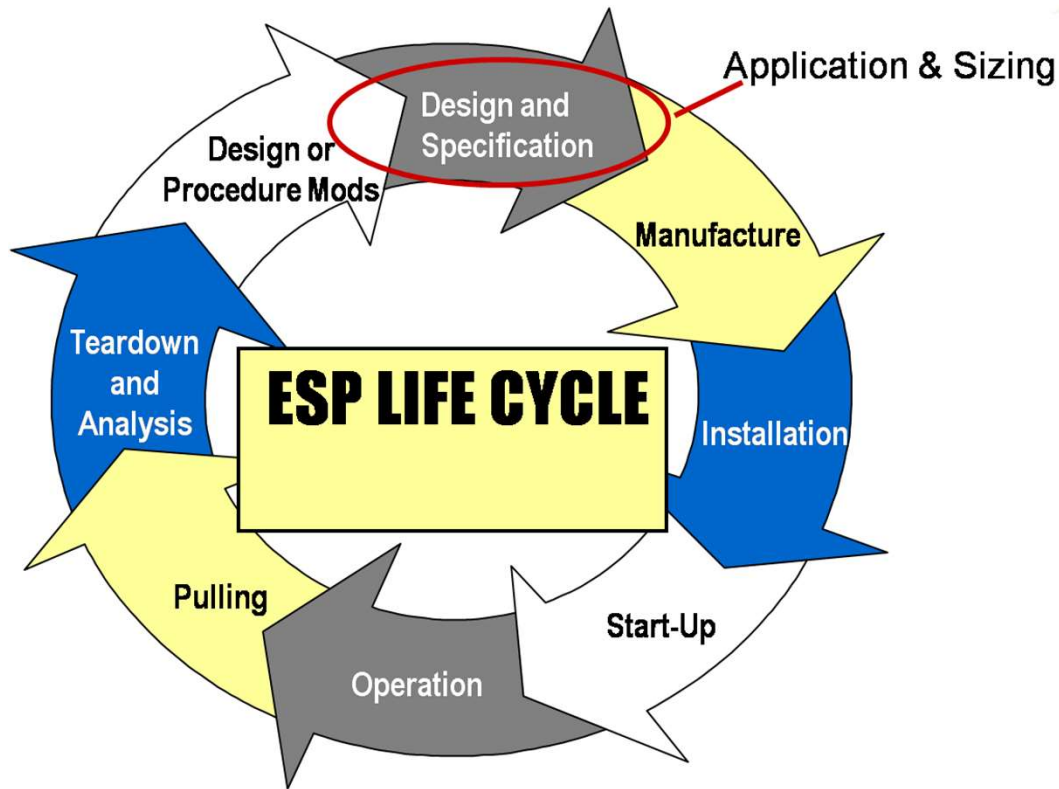
NAVIGATE From Junior To Leaders

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ESP Standard Sizing



Applications and Sizing



Learning objectives

1. Identify the definition and purpose of sizing
2. Define well data input parameters
3. Apply the productivity index formula
4. Compute maximum flow
5. Determine desired rates of flow and pressure
6. Demonstrate ability to convert a column of liquid to a pressure
7. Determine the specific gravity of a liquid
8. Determine the specific gravity of two combined liquids
9. Calculate Pump Intake Pressure
10. Calculate Total Dynamic Head
11. Select equipment suitable to a non-gas application

What is sizing ?

Designing the ESP system to be as **efficient**,
productive, and **reliable** as possible by
making sure it is correctly customized to the
conditions in a given well

Sequence

- **Step 1. • Collect and Analyze Data**
- Step 2. Determine Pump Intake Pressure
- Step 3. Gas Calculations
- Step 4. • Determine Total Dynamic Head
- Step 5. Select Pump Type
- Step 6. • Select Seal and Motor
- Step 7. • Select Power Cable
- Step 8. Select Motor Controller

Step 1. Collect and analyze data

Minimum Data for Proper ESP Sizing

- Well Identification
- Production Casing/Liner Profile
- Tubing Size, Weight and Connection
- Pump Setting Depth (*PD* and *PTVD*)
- Water Cut (*WC*)
- Producing Gas Ratio (*GOR* or *GLR*)
- Reservoir Temperature (*BHT*)
- Wellhead Tubing Pressure (*WHTP*)
- Pump Intake Pressure (*PIP*)
- Expected Production Rate (*Q*)
- Operating frequency (*Hz*)
- Maximum expected kill fluid density (*SG_{max}*)

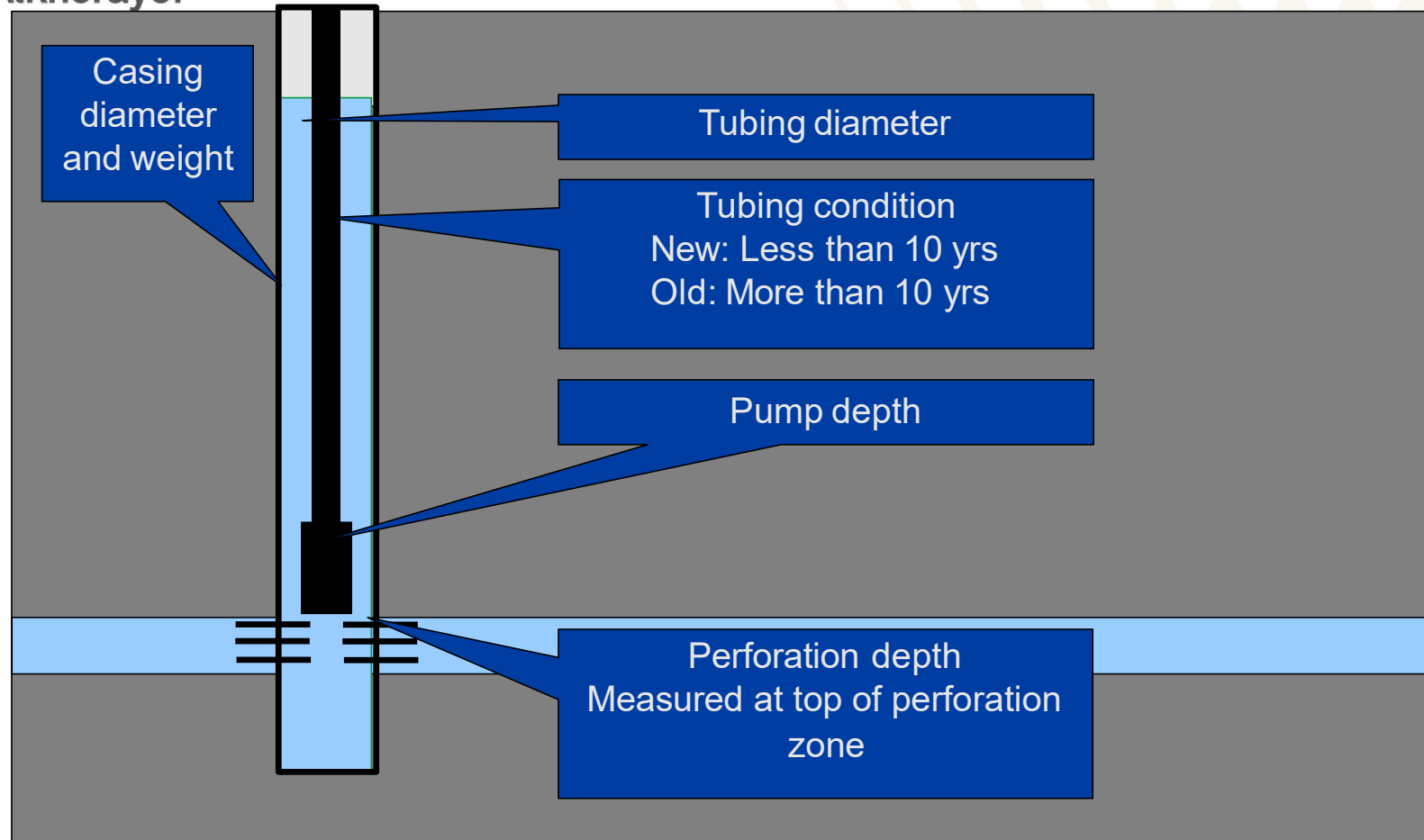
To Improve Design, Also Provide:

- Deviation Survey
- Oil, Water and Gas gravity
- Current Plugged Back Measured Depth (*PBTD*)
- Producing Interval
- Pressure Datum
- Static Bottom Hole Pressure (*SBHP*)
- Well Test production Rate (*Q_{test}*)
- Well Test Pressure (*TBHP*)
- Well test tubing and casing pressure (*WHTP* & *WHCP*)
- Gas separation efficiency
- Bubble point pressure (*P_b*)
- Solids Content
- Sour Gas Content

To Really “Nail” the Design, Also Provide:

- Percent Gas & Liquid Produced Up Casing
- Wellhead Temperature
- Separator Pressure and Temperature
- Gas Impurities
- Reservoir Type
- Best Friction Loss Correlation
- Best Inflow Performance Correlation
- Best PVT Correlations
- PVT and Viscosity Data
- Gas Analysis
- Water Analysis

Step 1. Well Data



1.1. Mechanical Data

- Casing size and weight
- Tubing size, weight and thread
- Well depth (both measured and true vertical)
- Perforation depth (both measured and true vertical)
- Unusual conditions such as tight spots, doglegs, liners and deviation from true vertical at desired setting depth.
- Well bore survey if the well is deviated or directional.

1.2. Production Data

- Current and desired production rate
- Oil production rate
- Water production rate
- GOR, free gas, solution gas, and gas bubble point
- Static BHP and fluid level
- Producing BHP and stabilized fluid level
- Bottom hole temperature
- System backpressure from flow lines, separator, and wellhead choke

1.3. Fluid Data

- Oil API gravity, viscosity, pour point, paraffin content, sand, and emulsion tendency
- Water specific gravity, chemical content, corrosion potential, and scale-forming tendency
- Gas specific gravity, chemical content, and corrosion potential
- Reservoir FVF, bubble point pressure, and viscosity/temperature curve.

1.4. Power Supply

- Primary grid voltage
- Primary grid frequency
- Capacity of the service
- Quality of service (spikes, sags, etc.)
- Power supply source (commercial grid, on site generator, shared generator, operator owned grid etc.)
- Any special requirements such as high ambient temperatures, hazardous locations etc.

Sequence

- Step 1. Collect and Analyze Data
- **Step 2. Determine Pump Intake Pressure**
- Step 3. Gas Calculations
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- Step 6. Select Seal and Motor
- Step 7. Select Power Cable
- Step 8. Select Motor Controller

Step 2. Determine pump intake pressure

- Pump Intake Pressure (PIP)
 - Natural pressure measurement
 - Pressure of column of fluid
 - Minimum requirements
 - Changes with flow
- To determine PIP
 - Composite Specific Gravity (SG_L)
 - Flowing Pressure (P_{wfd})
- To determine P_{wfd}
 - Productivity Index (PI)
- *To understand PI, we must first know about natural pressure...*

Step 2. Determine pump intake pressure

- **Natural pressure** (or reservoir pressure) refers to the amount of hydrostatic head a column of fluid has without any help from technology
- **Productivity index** compares two different flow and pressure tests to determine how much natural pressure will affect production
 - *PI represents the willingness of the formation to give up fluid*

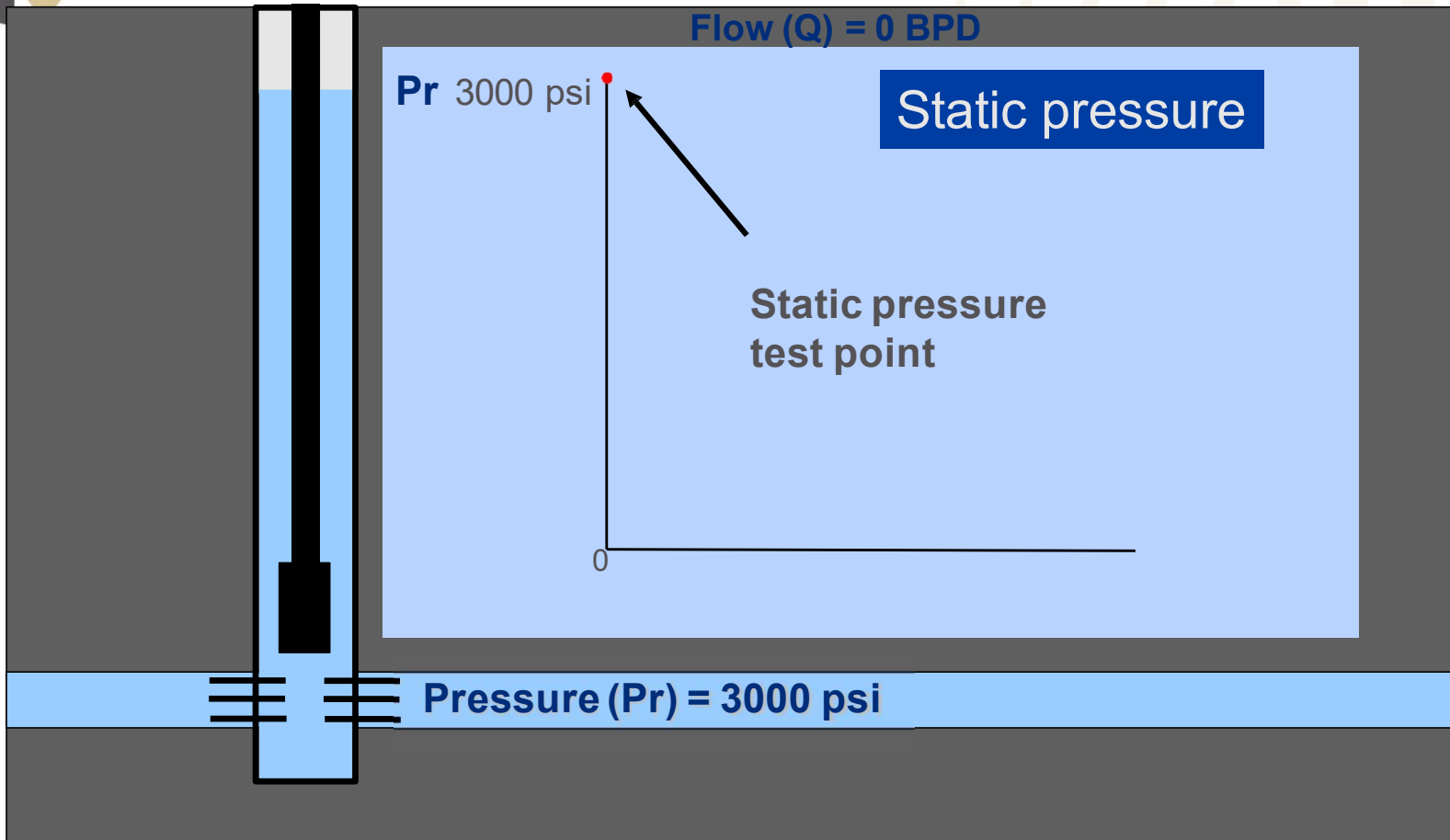
Step 2. Determine pump intake pressure

Productivity index

- Productivity Index (PI) compares *two flow and pressure tests*
- Flow
 - Represented by Q for quantity
 - Oil: Barrels Per Day (BPD)
 - Water: Gallons Per Minute (GPM)
- Pressure
 - Represented by P and can be measured in a flowing or static fluid
 - Pr: Reservoir pressure (static bottom-hole pressure)
 - Pwf: Pressure well flowing (also at bottom-hole)
 - Measured in Pounds per Square Inch (psi)

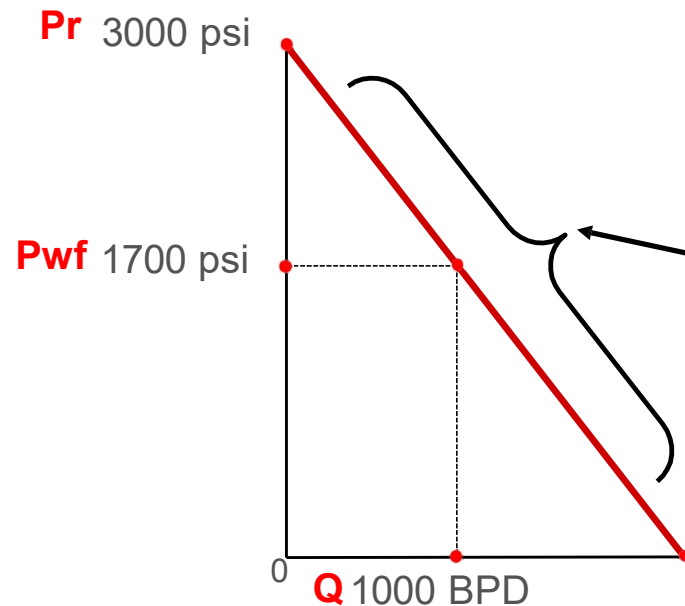
Step 2. Determine pump intake pressure

Productivity index



Step 2. Determine pump intake pressure

Productivity index



$$PI = \frac{Q}{P_r - P_{wf}}$$

$$PI = \frac{1000 \text{ BPD}}{3000 \text{ psi} - 1700 \text{ psi}}$$

$$= .77 \frac{\text{BPD}}{\text{psi}}$$

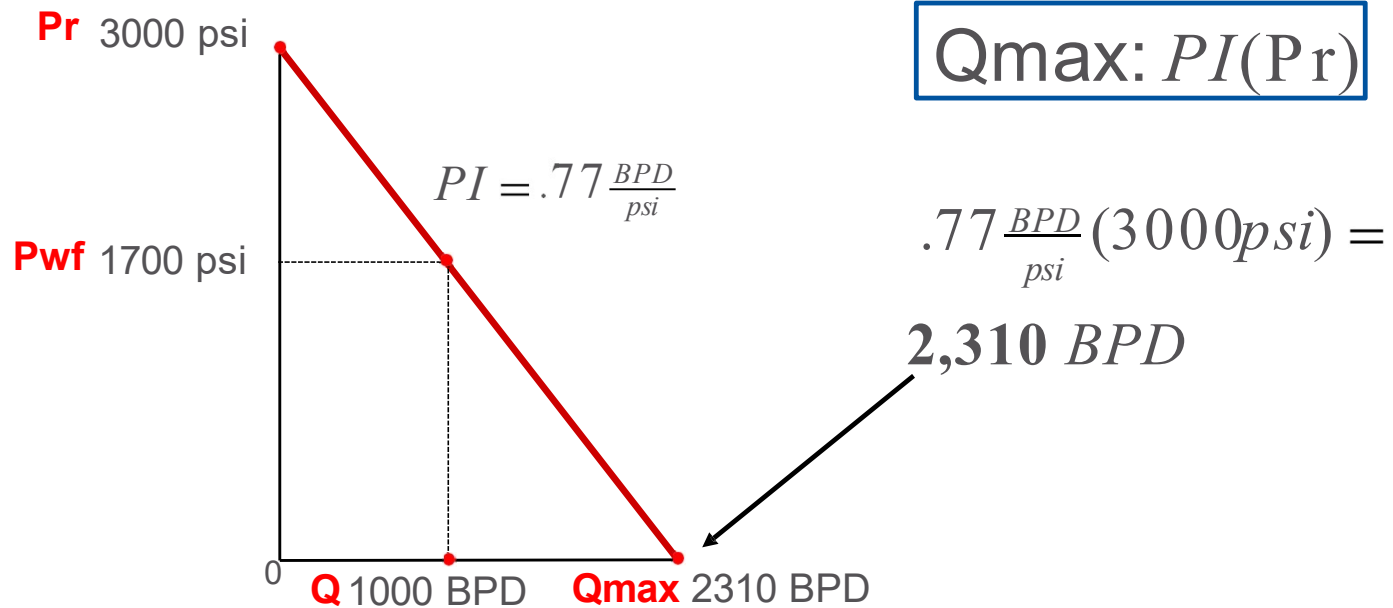
Maximum flow

- Qmax is the **theoretical** maximum flow possible from a well

– The formula is: $PI(P_r)$

– Qmax provides an upper flow limit when sizing the pump

Maximum flow





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Step 2. Determine pump intake pressure

Pressure well flowing, desired

- Pwfd is the pressure remaining (at datum depth) when producing a target volume of fluid

– *Pwfd is the pressure corresponding with a given rate of flow*

– The formula is:

$$P_r - \frac{Q_d}{PI}$$

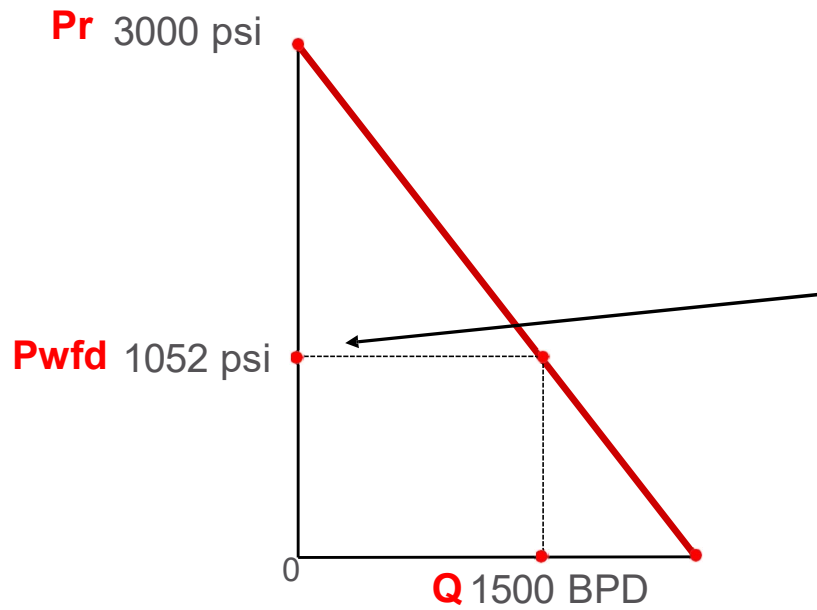
- The following example will determine the pressure remaining when producing 1500 BPD



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Step 2. Determine pump intake pressure

Pressure well flowing, desired



$$P_{wfd} = P_r - \frac{Q_d}{PI}$$

$$3000 \text{ psi} - \frac{1500 \text{ BPD}}{.77 \frac{\text{BPD}}{\text{psi}}} = 1052 \text{ psi}$$

Step 2. Determine pump intake pressure

Flow at a given pressure

– Variables previously solved:

- Productivity index (PI)
- Theoretical maximum flow (Q_{max})
- Pressure at a given flow (P_{wfd})

• *Flow at a given pressure (Q_d)* $Q_d = PI(P_r - P_{wfd})$

- How much fluid will be produced at a fixed (downhole) pressure?
- Example will determine flow rate with a downhole pressure of 200 psi

Step 2. Determine pump intake pressure

Flow at a given pressure



$$Q_d = PI(P_r - P_{wfd})$$

$$.77 \frac{\text{BPD}}{\text{psi}} (3000 \text{ psi} - 200 \text{ psi})$$

$$= 2156 \text{ BPD}$$

2. Determine pump intake pressure

Solving for reservoir pressure

- Reservoir Pressure - P_r

- Datum depth pressure with the well shut in (not flowing)

- Can be determined by comparing two flowing tests

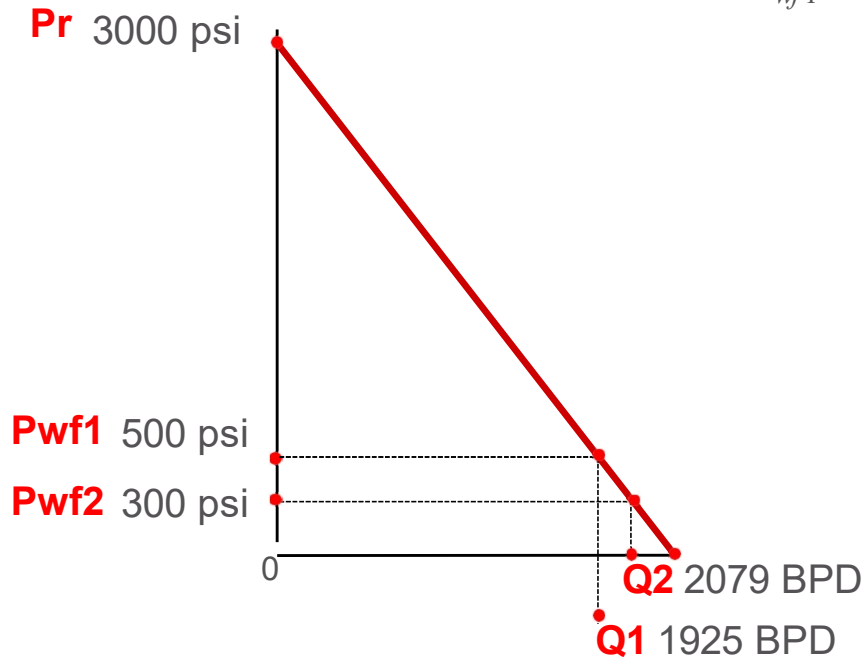
- The formula is:

$$P_r = \frac{Q_1}{PI} + P_{wf1}$$

Step 2. Determine pump intake pressure

Solving for reservoir pressure

$$PI = \frac{Q_2 - Q_1}{P_{wf1} - P_{wf2}} = \frac{2079 - 1925}{500 - 300} = .77 \frac{BPD}{PSI}$$



$$Pr = \frac{Q_1}{PI} + P_{wf1}$$

$$\frac{1925}{.77} + 500 \cong 3000 \text{ PSI}$$

Quick Review

We've learned about:

- Productivity Index (PI)
- Theoretical Maximum Flow Rate (Q_{max})
- Pressure at a Desired Rate of Flow (P_{wfd})
- Quantity at a Desired Pressure (Q_d)
- Solving for Reservoir Pressure from two flowing tests (P_r)

Quick Quiz

Flow is represented by Q for “quantity.” It’s usually measured in Barrels Per Day (BPD)

Pressure, measured in pounds per square inch (psi), is represented by P and can be measured in a flowing or static fluid

$$\underline{PI} = \frac{Q}{Pr - P_{wf2}}$$

$$\underline{P_{wfd}} = Pr - \frac{Qd}{PI}$$

$$\underline{Q_{max}} = PI (Pr)$$

$$\underline{Qd} = PI (Pr - P_{wfd})$$

Coming up ... specific gravity and head



Step 2. Determine pump intake pressure

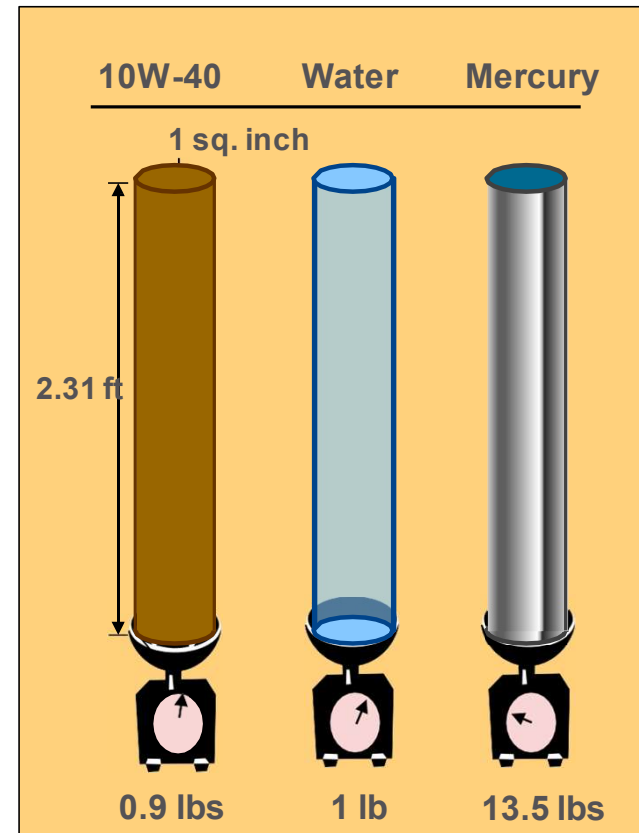
Specific gravity and head

- Two crucial factors to correctly sizing an ESP application
 - The density of the fluid being pumped (specific gravity)
 - The pressure exerted by a vertical column of fluid (head)

Step 2. Determine pump intake pressure

Specific gravity and head

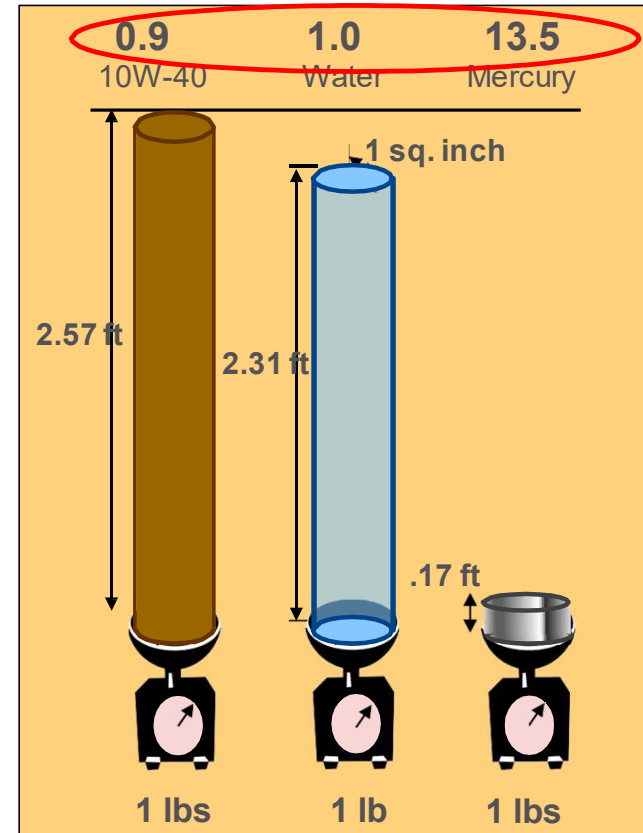
- A column of fluid exerts pressure at the bottom
- Specific gravity determines how much pressure the column will exert
- Specific gravity is a measurement of a material's density
- Benchmark is water (1.0)



Step 2. Determine pump intake pressure

Specific gravity and head

- A one pound column of water with a one square inch cross section is 2.31 feet high
- One psi of water equates to 2.31 feet of head
- Head and specific gravity (SG) correspond for all liquids
- A liquid with a SG lower than water will form a taller column of fluid
- A liquid with a SG higher than water will form a shorter column of fluid



Step 2. Determine pump intake pressure

Specific gravity and head

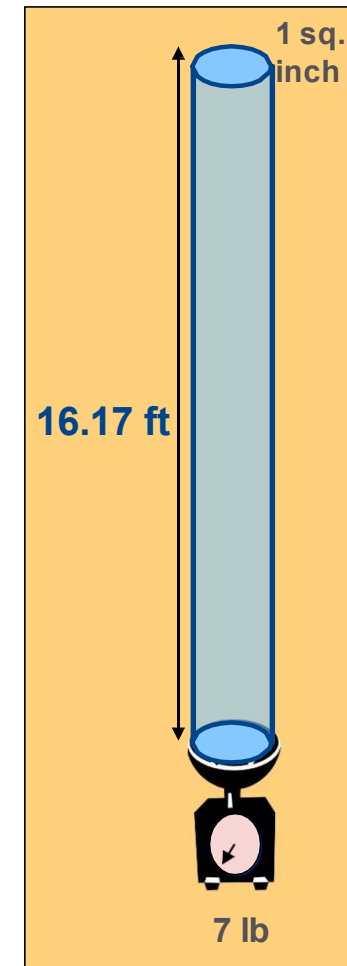
- Two formulas for determining head:

$$Head = \frac{psi(2.31 \frac{ft}{PSI})}{SpGr} = \frac{psi}{.433 \frac{psi}{ft} (SpGr)}$$

- Both formulas have the same result
- 7 psi of water...

$$Head(ft) = \frac{7 psi}{.433 \frac{psi}{ft}} =$$

$$Head(ft) = \frac{7 psi}{.433 \frac{psi}{ft}} =$$



Step 2. Determine pump intake pressure

Specific gravity and head

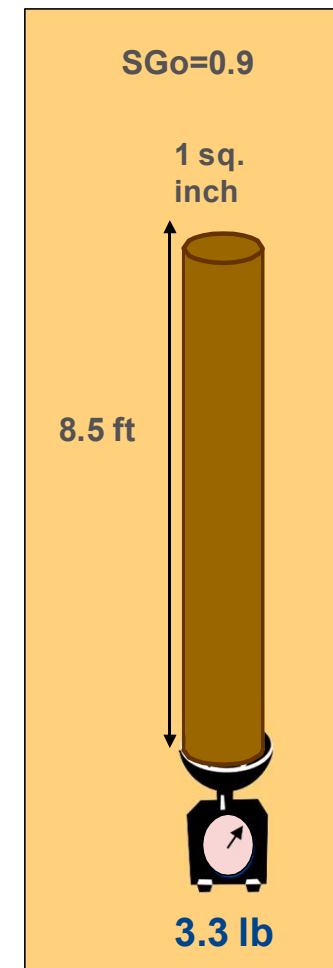
- Two formulas for determining *psi*

$$psi = \frac{Head(SpGr)}{2.31 \frac{ft}{PSI}} = Head(SpGr)(.433 \frac{PSI}{ft})$$

- Both formulas have the same result
- 8.5 feet of oil...

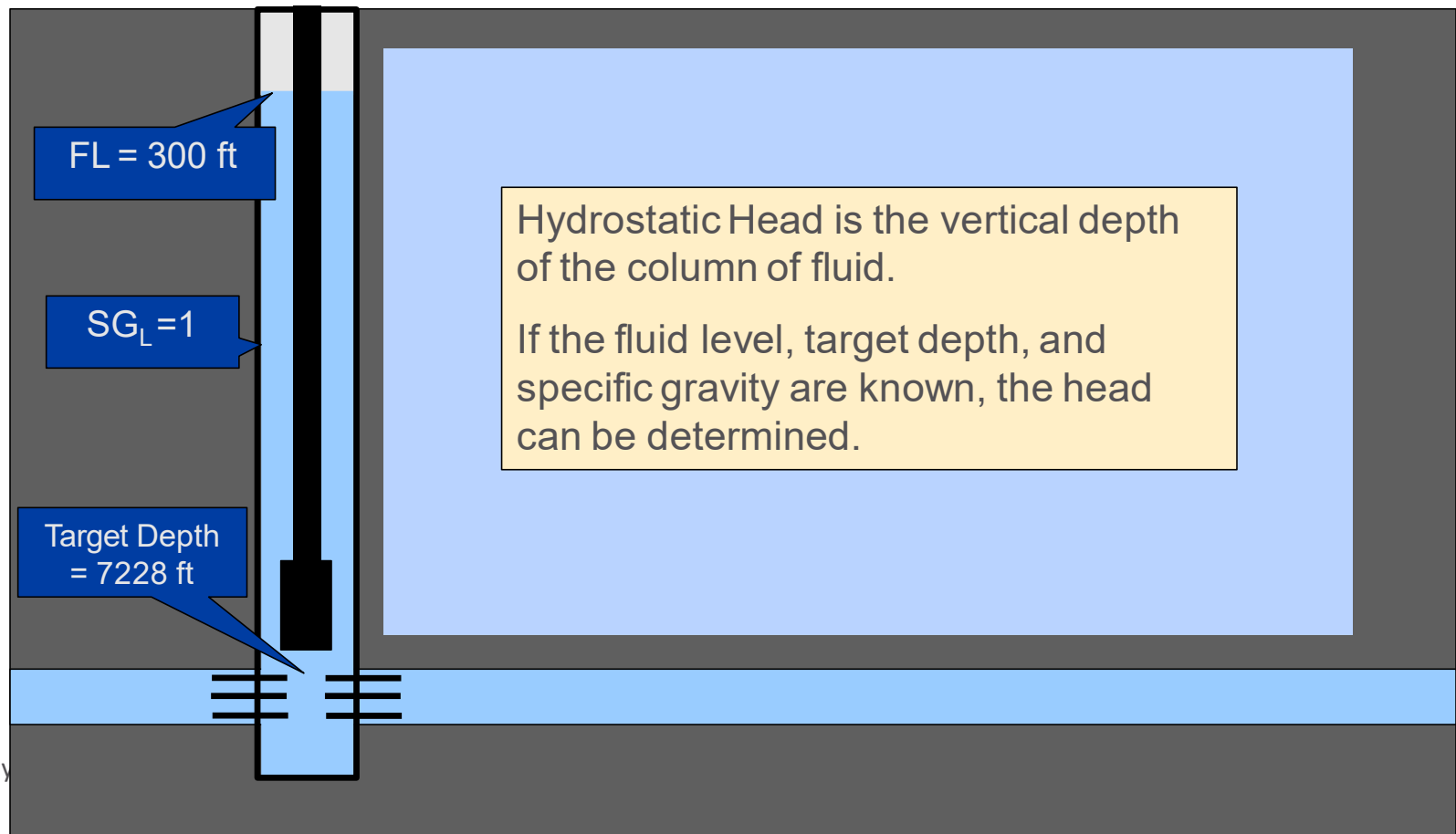
$$psi = \frac{8.5 \times 0.9 \times 62.4}{2.31} \approx$$

$$psi =$$



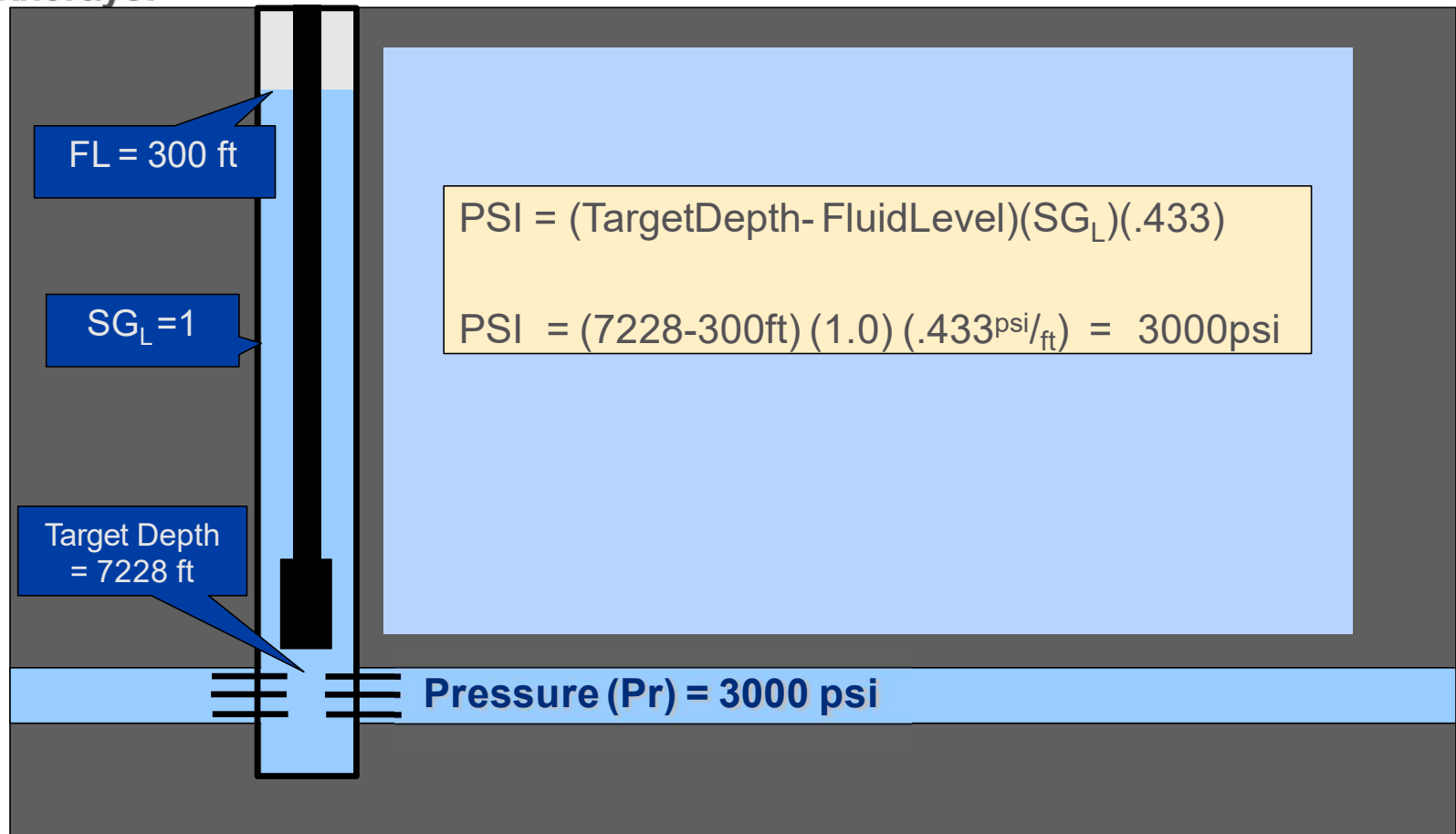
Step 2. Determine pump intake pressure

Computing pressure from fluid level data



Step 2. Determine pump intake pressure

Computing pressure from fluid level data



Example

Step 1: Collect and analyze data

- Well Data

- Casing – 7", 20lb/ft
- Tubing – 2 7/8", new
- Perforations – 5300 to 5400 ft
- Pump setting depth – 5200 ft
- Bottom-hole temp – 180° F

- Fluid Conditions

- SG_W – 1.02
- SG_O – 0.876
- SG_G – NA
- Bubble point – NA

Production Data

- Surface pressure – 150 psi
- Test rate (Q) – 900 bpd
- Test Pressure (P_{wf}) – 985 psi
- Datum depth – 5350 ft
- Natural pressure (P_r) – 1650 psi
- Gas-oil ratio – unknown
- Water cut – 90%
- Desired rate (Q_d) – 2000 bpd

Power Source

- Available voltage – 12,470 V, 3 Phase
- Frequency – 60 Hertz



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Step 2: Determine pump intake pressure

Using the Productivity Index (PI) method find:

- Productivity Index (PI)
- Pressure Well Flowing Desired (P_{wfd})
- Composite Specific Gravity (SG_L)
- Pump Intake Pressure (PIP)

Step 2: Determine pump intake pressure

Step 2-1. Solve for PI

$$PI = \frac{900 - 0}{(1650 - 985)} = 1.353 \frac{bpd}{psi}$$

Known Variables:

$$Q1 = 900 \text{ bpd}$$

$$Q2 = 0 \text{ bpd}$$

$$Pr = 1650 \text{ psi}$$

$$Pwf1 = 985 \text{ psi}$$

$$Qd = 2000 \text{ bpd}$$

Step 2-2. Solve for Pwfd at the desired production rate

$$Pwfd = Pr - \frac{Qd}{PI} = 1650 - \frac{2000}{1.353} = 172 \text{ psi}$$

Step 2: Determine pump intake pressure

Step 2-3. Determine composite specific gravity (SGL)

Known Variables:

SpGrW – 1.02

SpGrO – 0.876

Water Cut – 90% (= 0.9)

$$SG_L = SG_W(WC) + SG_O(1 - WC)$$

$$SG_L = \quad + \quad =$$

If needed, use the following formula to convert from API gravity to specific gravity:

$$SG_O = \frac{141.5}{API_O + 131.5}$$

Step 2: Determine pump intake pressure

Step 2-3. Determine pump intake pressure (PIP)

Known Variables:

$$SG_L = 1.01$$

$$\text{Datum} = 5350 \text{ ft}$$

$$\text{P-depth} = 5200 \text{ ft}$$

$$P_{wfd} = 172 \text{ psi}$$

$$PIP = P_{wfd} - \left[\frac{(\text{Datum} - \text{Pdepth}) SG_L}{2.31 \frac{\text{ft}}{\text{psi}}} \right]$$

$$PIP = - \left[\frac{\text{Datum} - \text{Pdepth}}{2.31} \right] =$$

Sequence

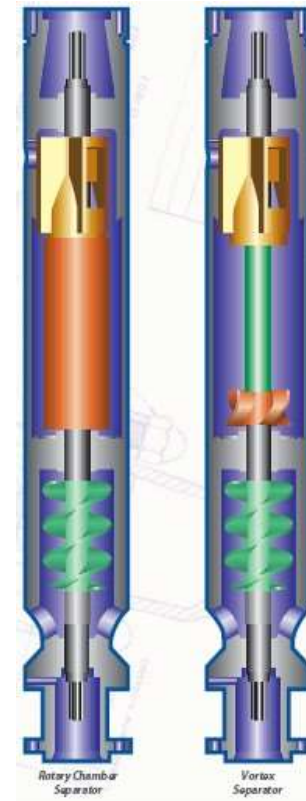
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Step 3: Gas calculations

- Gas significantly effects the performance of an ESP
- The percentage of free gas is essential to designing the system, and is usually available in the well data
- Example assumes gas will remain in solution

Step 3: Gas calculations

- Gas occupies space and the pump
- Pump can not tell the difference in gas or fluid as they both occupy space
- The amount of free gas must be calculated if unavailable
- Pumps can handle 10% - 15% of free gas without significant problems



Step 3: Gas calculations

STANDARD PUMP
Radial Impeller



STANDARD PUMP
Mixed Flow Impeller



TAPERED



PUMP W/SEPARATOR



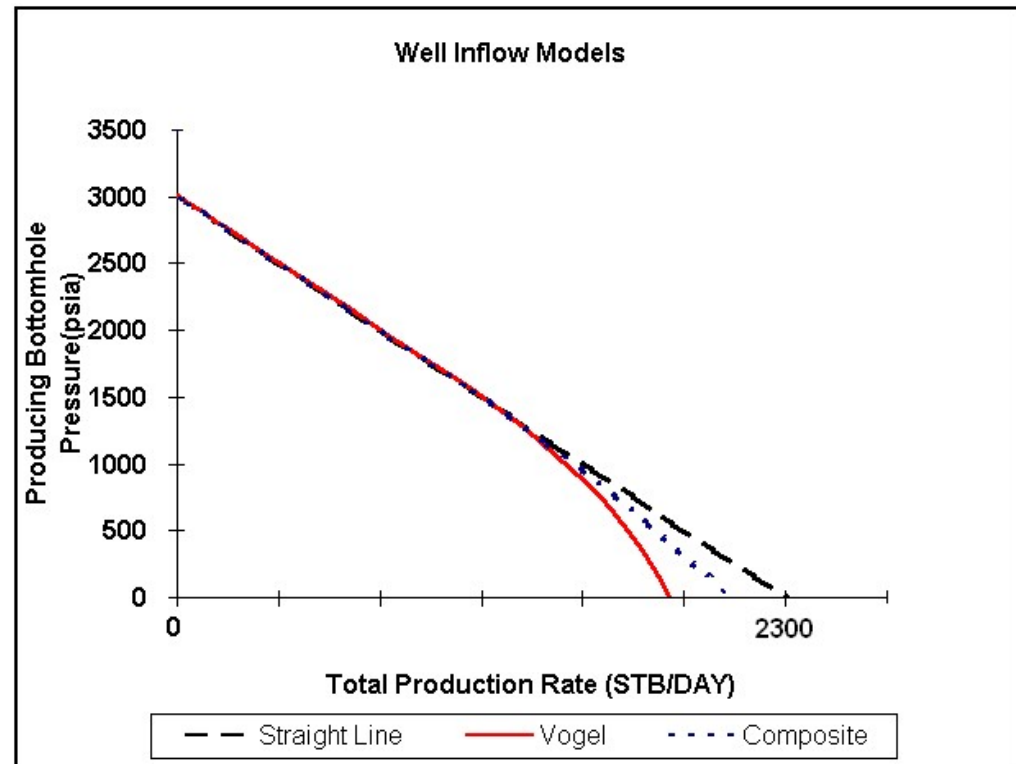
TAPERED PUMP
W/SEPARATOR



ABILITY TO HANDLE FREE GAS

Step 3: Gas calculations

- Gas calculation
 - Vogel (Inflow performance Relationship)
 - Gassy applications
 - Low water cut
 - Productivity Index (PI)
 - No gas present
 - 100% water cut
 - Composite
 - Some gas present
 - Oil water mixture



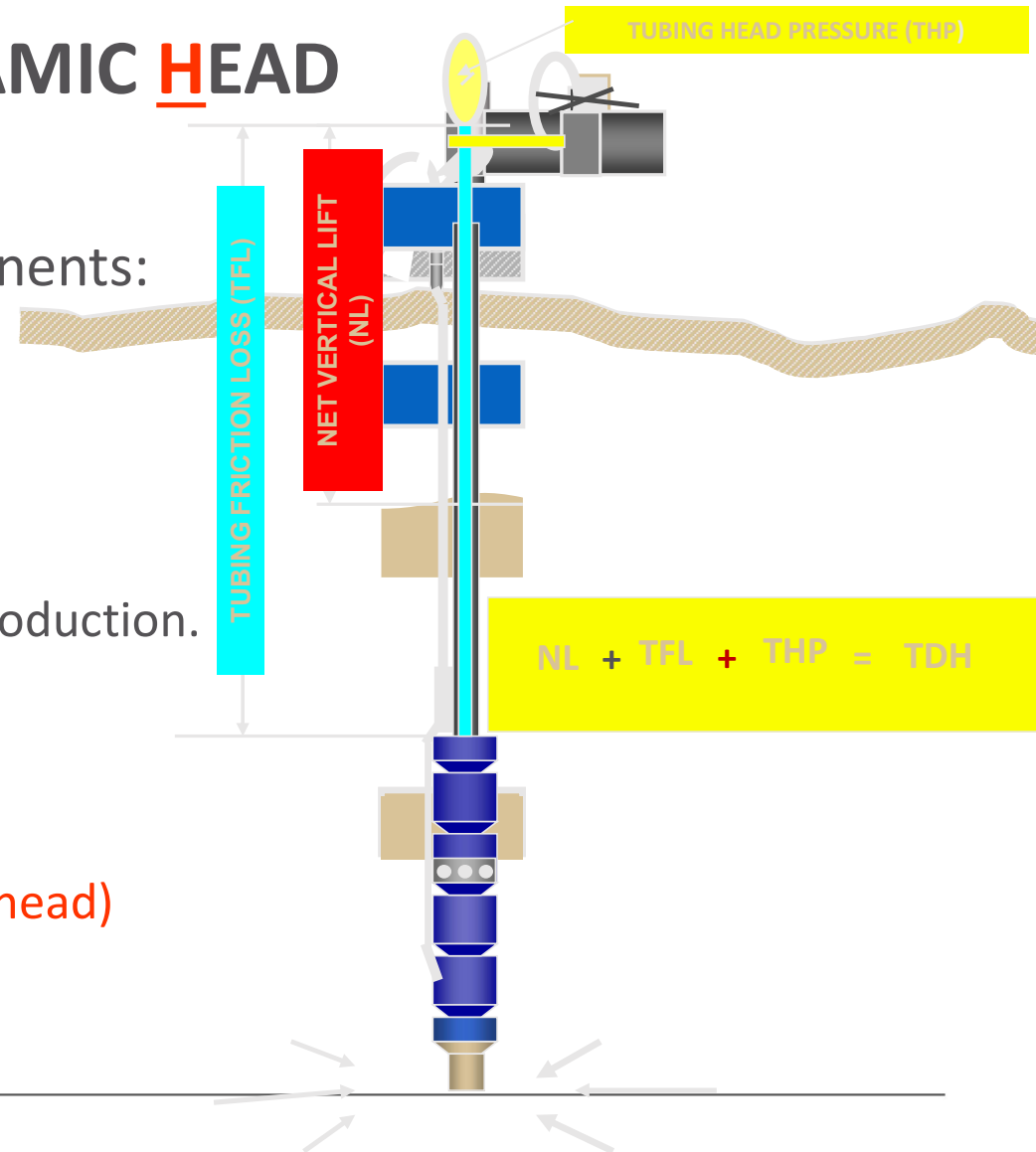
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TOTAL DYNAMIC HEAD

TDH is the sum of three basic components:

- 1) Net Vertical Lift (NL) =
is the net distance where the fluid must be lifted
- 2) Friction loss (TFL) =
Flow disturbance in the tubing string during production.
- 3) Tubing Head Pressure (THP) =
Pressure which the unit must pump against
(back pressure / flow line pressure at the well head)



“**Net Vertical Lift**” is the vertical distance in the empty space in the tubing where the fluid must be lifted to get to the surface.

The energy required to lift the fluid can be determine by the equation :

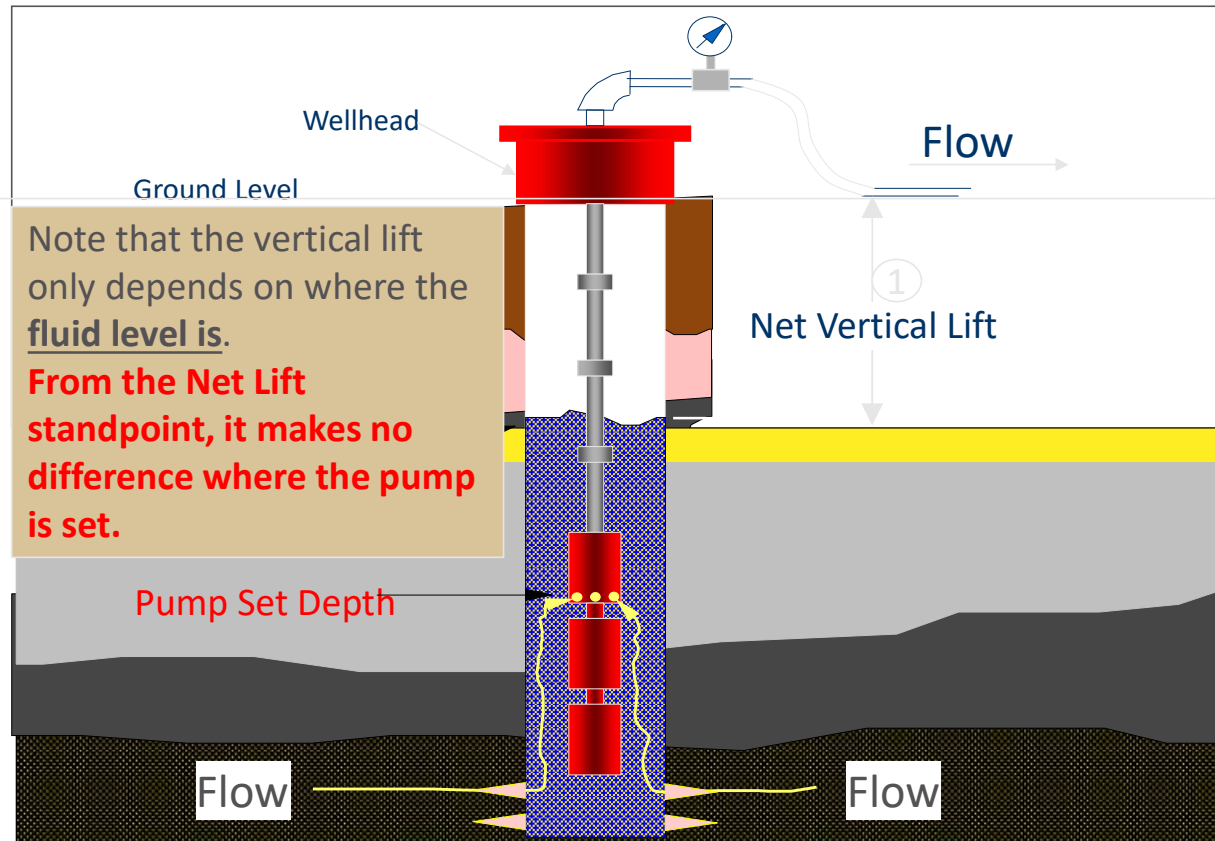
$$\text{Work} = mg\Delta h$$

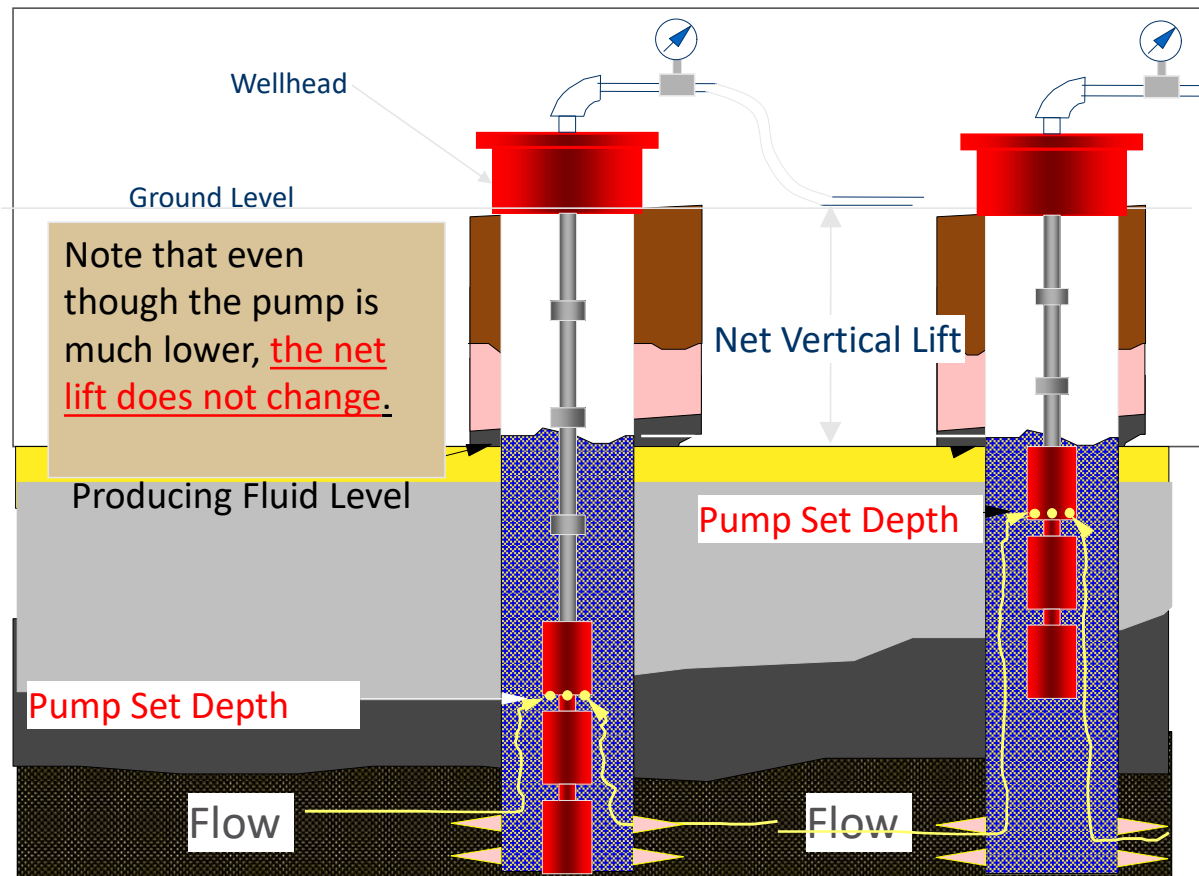
Where:

m = mass of fluid

g = acceleration due
to gravity

Δh = height of the fluid is
lifted





What if, instead of lifting the fluid vertically, we move it sideways? How much work did we do?

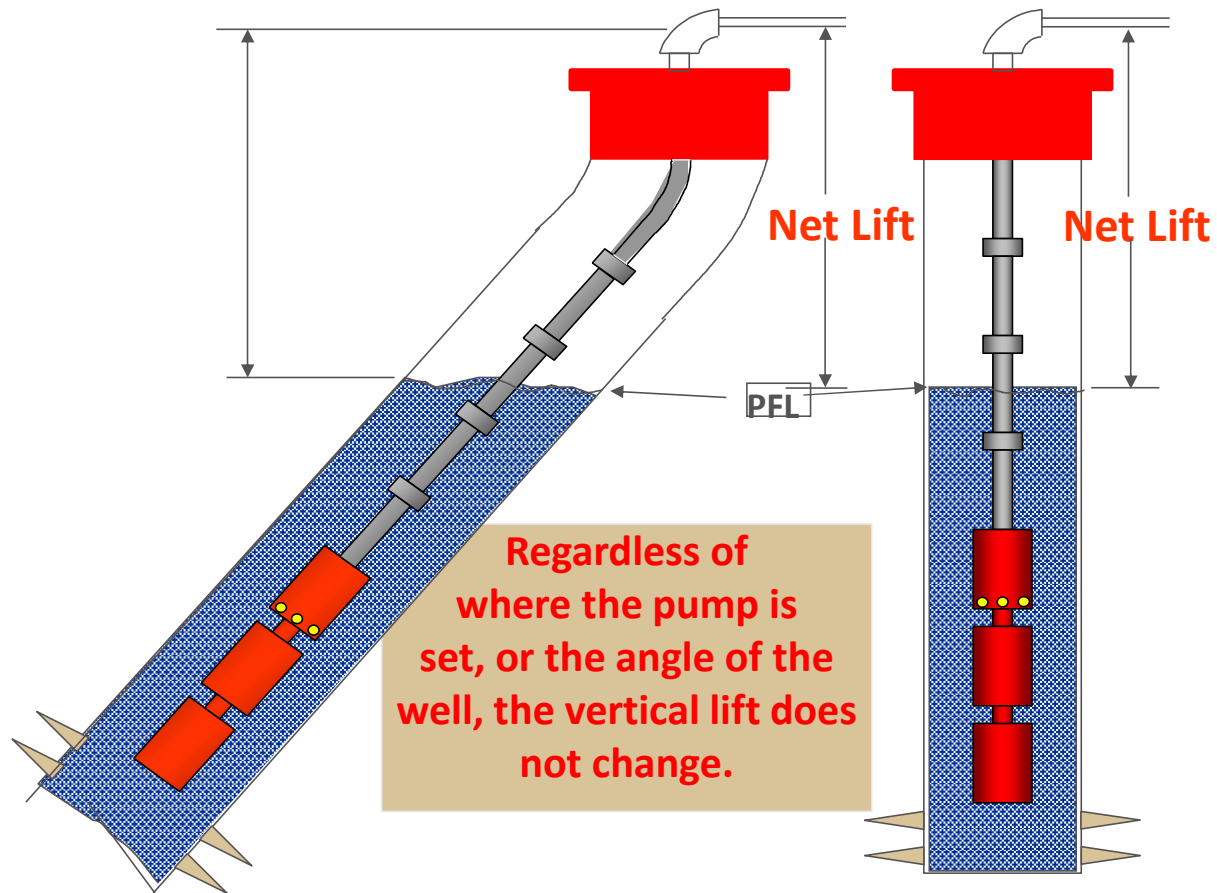
NONE !

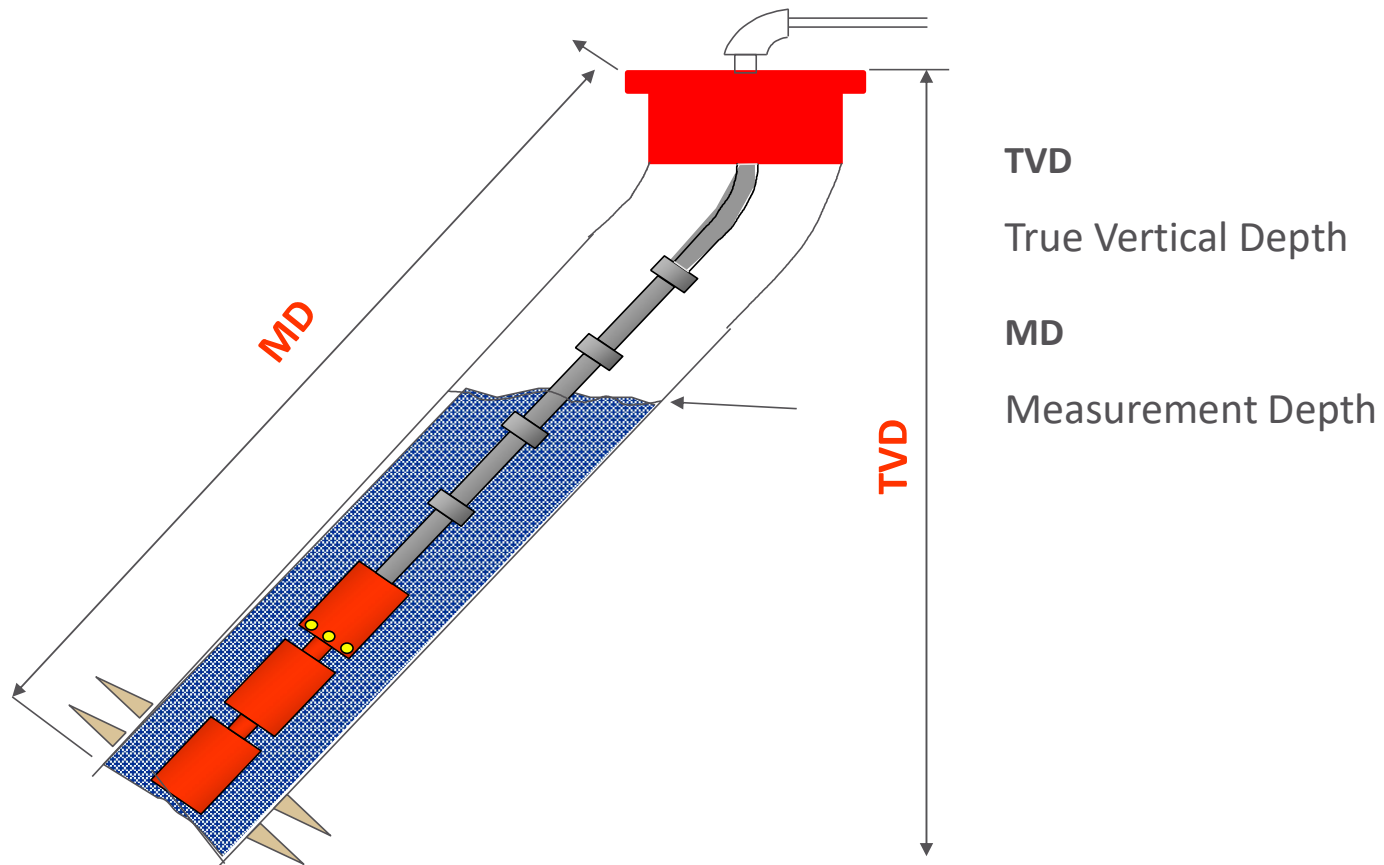
If Work (energy) = $mg\Delta h$,

On sideways moving $\Delta h = 0$, so the work must be zero.



What about deviated wells?





Example = assume a given fluid level of 4000 feet (TVD) from surface so:

$$\text{Net Vertical Lift} = 4000 \text{ ft}$$

Remember if the well is deviated, the total measured distance (MD) from surface could be much greater but, since the work done in moving the fluid sideways is zero, only the vertical distance is matters.

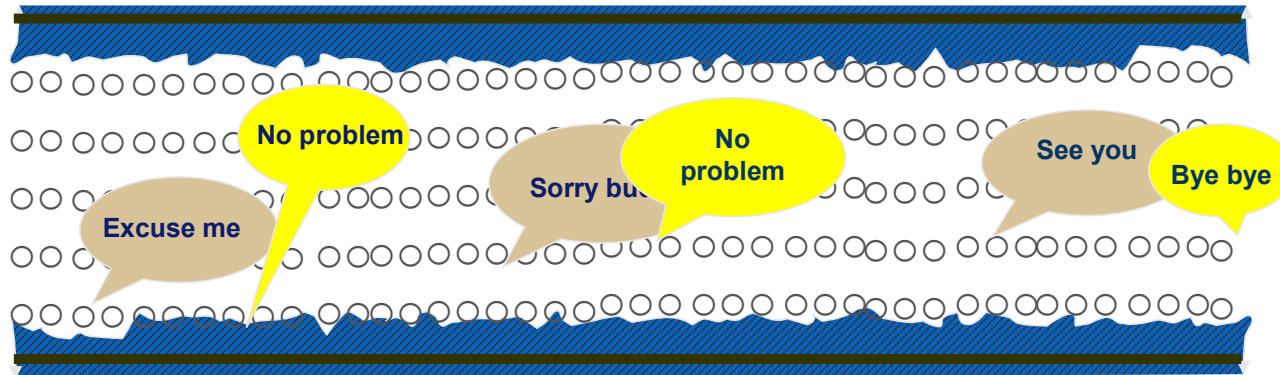
What is friction ?

Friction is an energy loss due to viscous shear of the flowing fluid.

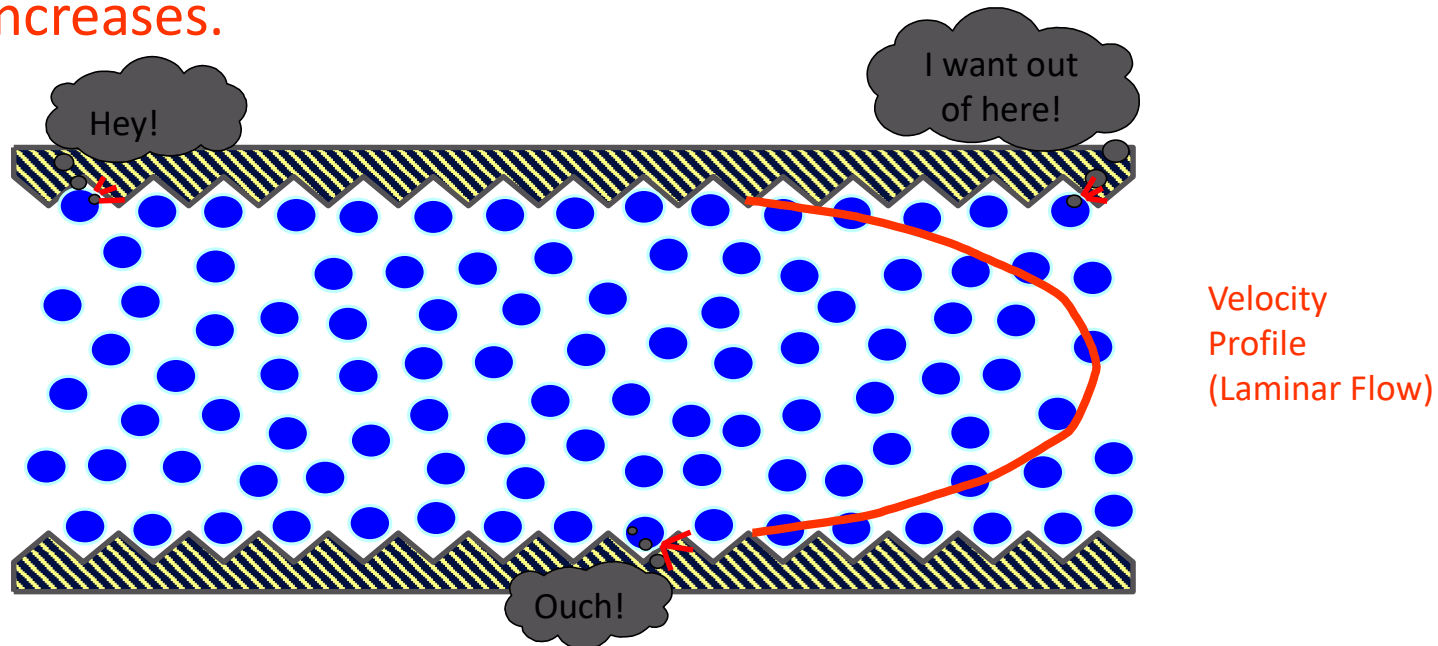
In a fluid, molecules are free to move past each other but there may be a little resistance. This resistance is due to shear forces which must be overcome.

The higher the shear forces, the higher the viscosity.

In a **single-phase** fluid, most of the liquid is moving along together so there is not much shear in the liquid itself and this friction can usually be ignored.



The walls of the pipe, however, will tend to "stick" to the fluid so shear forces between the pipe and the fluid can be quite large and **increase as the velocity of the fluid increases.**



The amount of friction can be represented by “friction factor” - (*f*)

Given “*f*”, we can calculate pressure lost :

$$\Delta P = \frac{f \rho v^2}{2 g_c d}$$

Where : ΔP = pressure loss ρ = fluid density

v = fluid velocity g_c = gravity constant

d = tubing diameter

What is the friction loss at zero flow?

At zero flow, $v = 0$, so the ΔP due to friction is zero.

$$\Delta P = \frac{f \rho v^2}{2 g_c d}$$

Assume the flow rate is not zero but is some constant value. What happens to the friction as the pipe diameter increases?

.....remember

$$\Delta P_{\downarrow} = \frac{f \rho v^2}{2 g_c d^{\uparrow}}$$

As the pipe diameter **increases**, the ΔP **decreases** as can be seen in the equation. But something else also happens.

What is it ?

As the pipe diameter increases, the velocity (**v**) decreases by the square of the diameter change so it is reduced drastically.

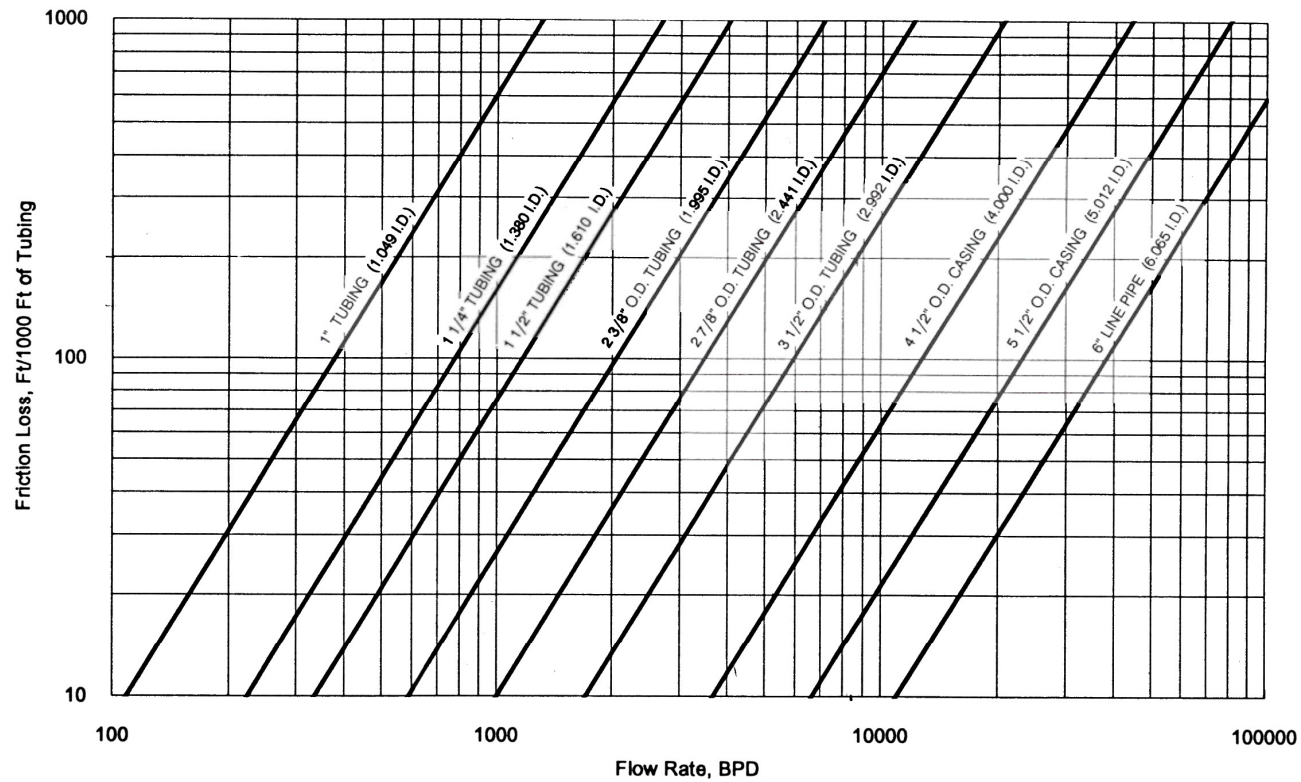
These two factors, make an increase in pipe diameter have a large impact on decreasing the **frictional pressure** losses.

How to calculate friction loss

Fortunately, there are many charts available for determining friction as we do not need to use these equations. A very useful chart for our purposes follows:



Friction Losses In A.P.I. Tubulars



Based on Hazen Williams Formula:
 $F = 2.083(100/C)^{1.85} (Q/34.3)^{1.85}/ID^{4.8655}$

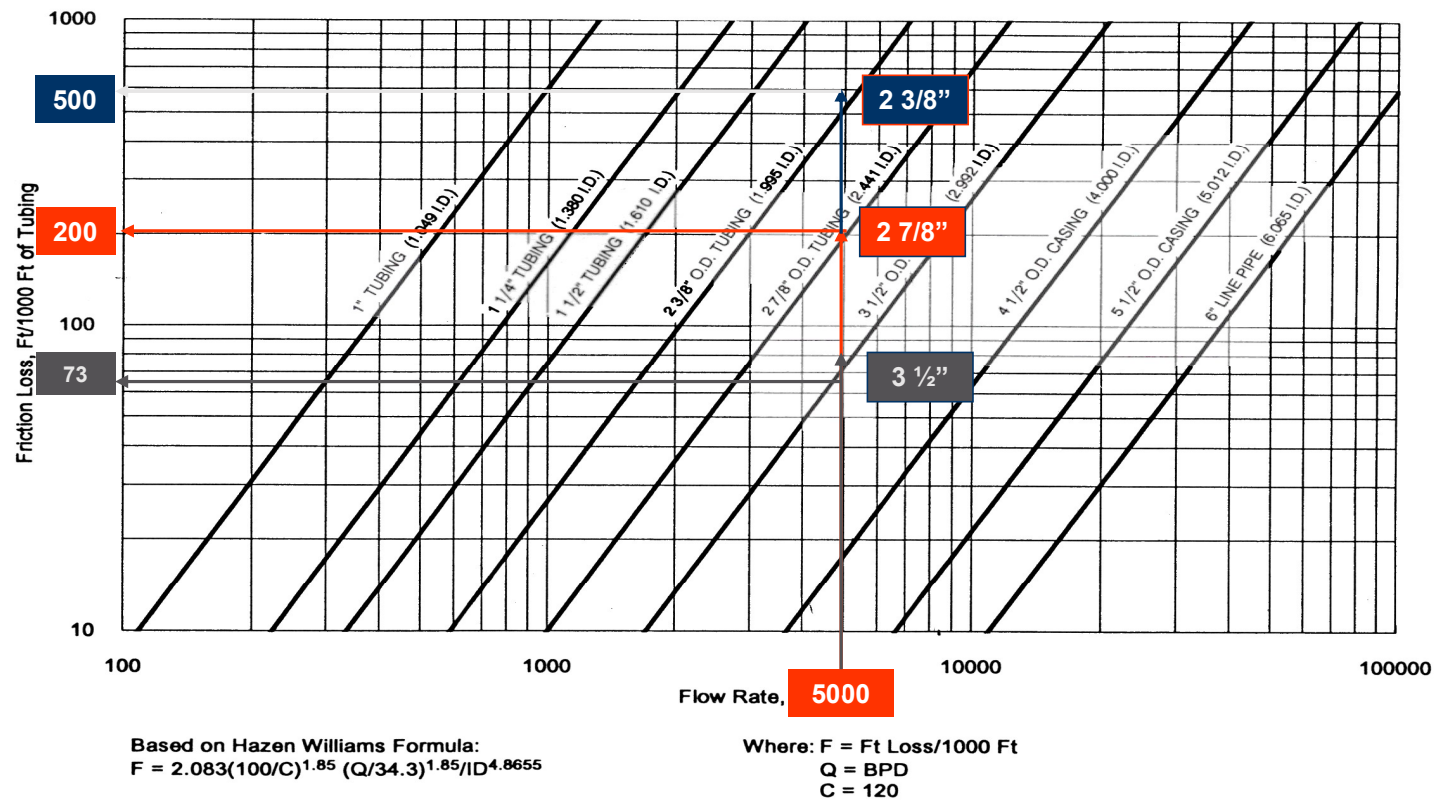
Where: F = Ft Loss/1000 Ft
 Q = BPD
 C = 120



How to use the chart:

Say, for example, we have a total tubing length (**pump setting depth**) of 6500 feet and we want to produce 5000 bpd. In this case we have three size of tubing size $2 \frac{3}{8}$ " , $2 \frac{7}{8}$ " and $3 \frac{1}{2}$ " available in stock. What will the friction be?

Friction Losses In A.P.I. Tubulars



Since we have 6500' of tubing,

Total Friction Loss on :

- 2 3/8" tubing = $500 \times (6500 \text{ Ft} / 1000) = 3200 \text{ Ft}$
- 2 7/8" tubing = $100 \times (6500 \text{ Ft} / 1000) = 650 \text{ Ft}$
- 3 1/2" tubing = $73 \times (6500 \text{ Ft} / 1000) = 474.5 \text{ Ft}$

Option ????????

Is bigger tubing always better?

NO

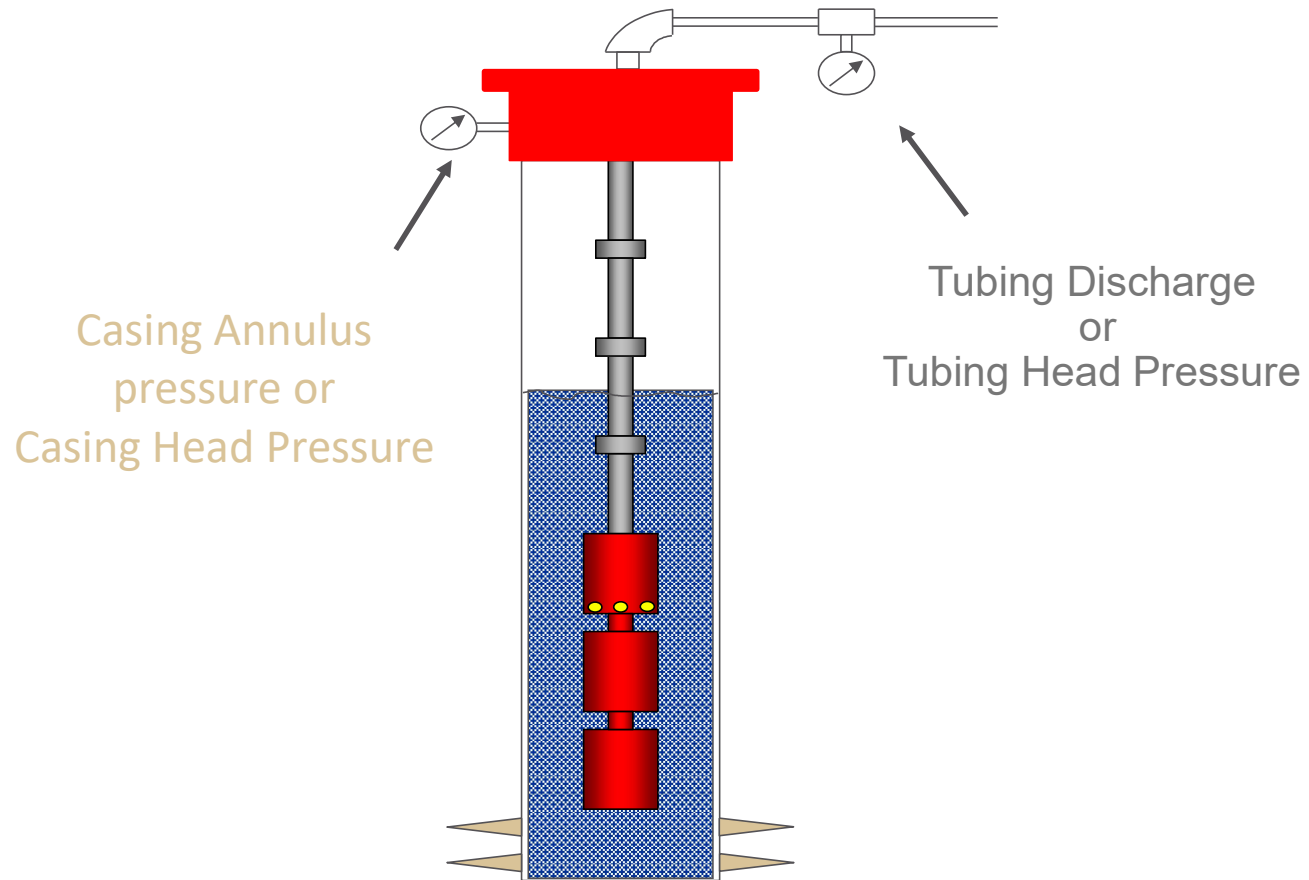
...potential problems due to solids in suspension (sand).

In the job site wellhead pressure is also called:

- Tubing Head Pressure (THP)
- Surface Pressure
- Back Pressure
- Flow line Pressure
- Discharge Pressure
- Well head Pressure

Technically there are two "Wellhead Pressures".

What are they ?





Up to this point, we have been calculating everything in terms of "feet". This is very convenient when sizing a pump.

For example, given:

Well head pressure = 200 psi

Water Cut (1.07 sp. Gr.) = 60%

Oil Cut = ?

Assumed oil API = 30 °API

Determine oil SG, use the API gravity equation :

$$Sp.Gr. = \frac{141.5}{131.5 + API}$$

$$API = \frac{141.5}{Sp.Gr.} - 131.5$$

Determine oil SG, use the API gravity equation :

$$Sp.Gr. = \frac{141.5}{131.5 + API}$$

$$\text{Oil SG} = \frac{141.5}{131.5 + 30} = 0.876$$

Next, find the “composite” specific gravity of the fluid in the well?

The best way to do this is simply to take an "arithmetic average".

$$Sp. Gr. = (f_w \times \gamma_w) + (f_o \times \gamma_o)$$

Where :

f_w is the water fraction

γ_w is the water specific gravity

f_o is the oil fraction

γ_o is the oil specific gravity

$$\text{Sp. Gr.} = (f_w \times \gamma_w) + (f_o \times \gamma_o)$$

$$\text{Average SG} = (0.60 \times 1.07) + (0.40 \times 0.876) = 0.992$$

You have now average SG of the fluid so you are now ready to convert the wellhead pressure from [psi] to feet.

..... Remember equation given :

$$\text{Wellhead "Feet"} = \frac{\text{Wellhead Pressure} \times 2.31}{\text{sp.gr.}}$$

Using the numbers from the example:

$$\text{Wellhead "Feet"} = \frac{200 \text{ psi} * 2.31 \text{ ft/ psi}}{0.992} = 465 \text{ ft}$$

Wellhead (feet) also can be determined by equation:

$$\text{Wellhead (Feet)} = \frac{\text{Well head pressure}}{(\text{SG} * 0.433)}$$

$$\text{Wellhead (feet)} = \frac{200 \text{ psi}}{(0.992 * 0.433 \text{ psi/ ft})} = 465 \text{ ft}$$

→ **Fluid Gradient**

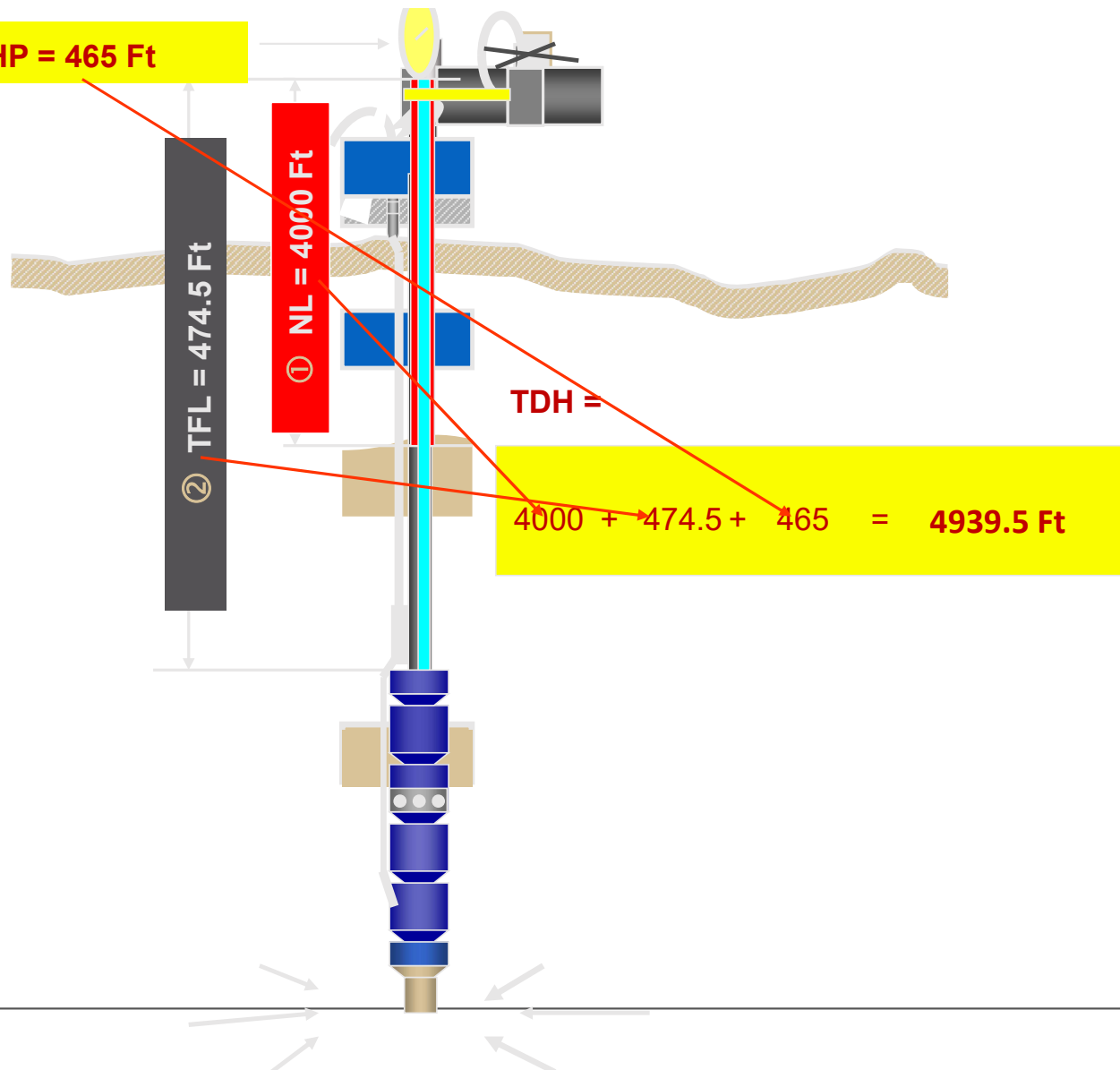
$(\text{SG} * 0.433)$



The TDH will be the sum of:

Net Lift,
Friction Loss, and
Wellhead pressure.

We will assume 3 1/2 tubing since it provide lowest friction



Sequence

- Step 1. • Collect and Analyze Data
- Step 2. • Determine Pump Intake Pressure
- Step 3. • Gas Calculations
- Step 4. • Determine Total Dynamic Head
- **Step 5. • Select Pump Type**
- Step 6. • Select Seal and Motor
- Step 7. • Select Power Cable
- Step 8. • Select Motor Controller

Step 5. Select pump type

APC Pump Specifications @ 60HZ

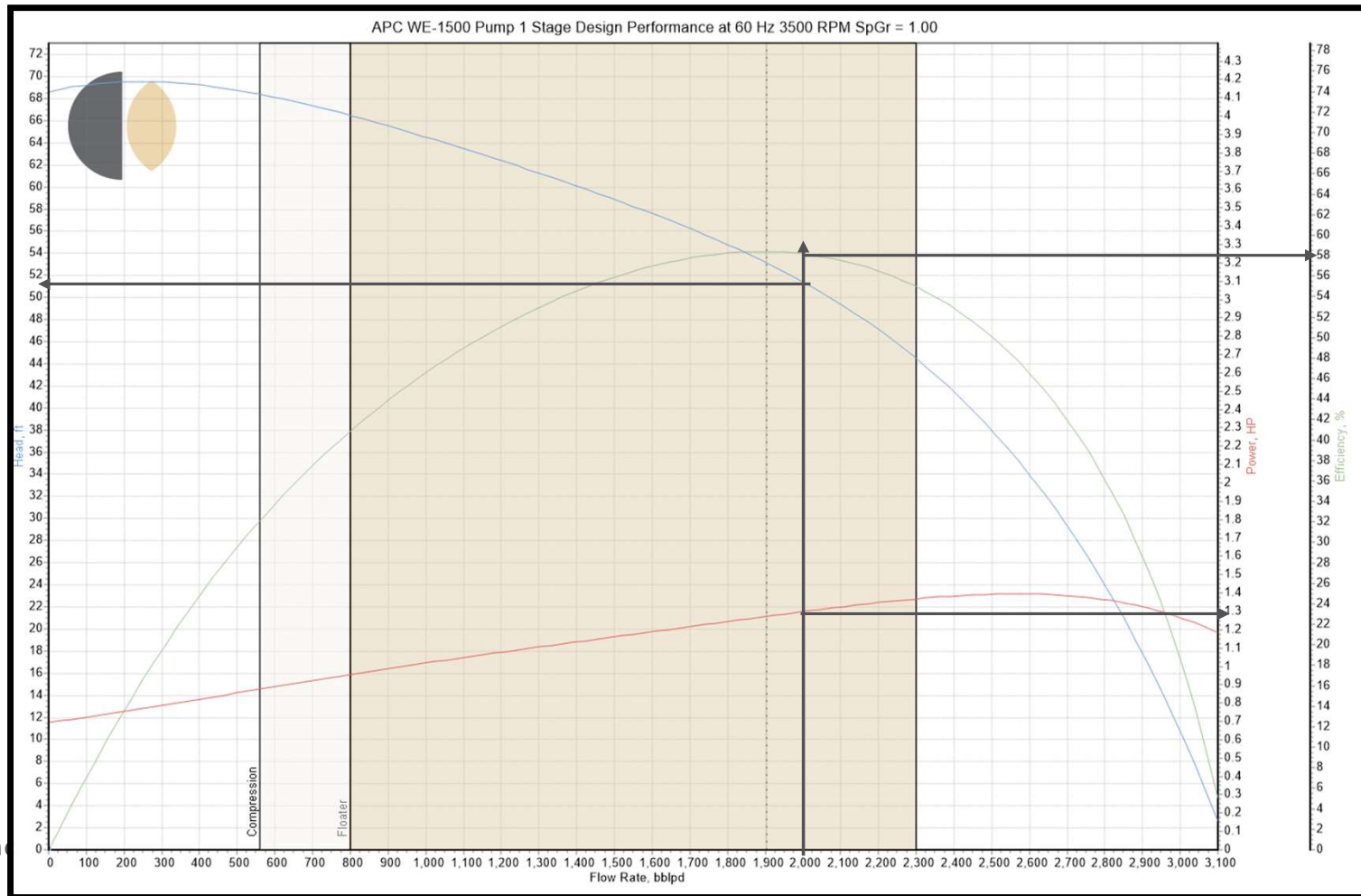
60 Hz (3500 RPM)												
Series	Type	OD (In)	Recommended Operating Range (BFPD)				Max HP per Stage	HP per Stage @ BEP	Head (ft) per Stage @ BEP	Max Head (ft) per Stage	Stage	
			Min		BEP	Max					Radial Flow	Mixed Flow
			FLT	CP								
338	WA-550	3.38	400	200	552	700	0.14	0.13	17.19	22.60	X	na
	WA-900		660	350	922	1060	0.17	0.17	13.59	21.79	X	na
	WA-1200		900	450	1160	1500	0.22	0.22	14.43	22.14	X	na
	WA-1500		1000	500	1630	2000	0.35	0.33	12.90	18.99	X	na
WD												
400	WD-150	4.00	90	90	231	280	0.08	0.08	17.89	23.41	X	na
	WD-450		250	200	475	550	0.12	0.12	16.37	27.92	X	na
	WD-650		450	250	683	800	0.20	0.20	23.44	33.20	X	na
	WD-850		550	300	838	1020	0.31	0.29	28.16	33.76	X	na
	WD-1000MF		700	200	1160	1300	0.41	0.41	28.22	39.60	na	na
	WD-1750		1100	400	1928	2150	0.42	0.42	21.76	33.42	na	na
	WD-3000		2000	900	2880	3700	0.63	0.62	18.57	28.03	na	na
	WD-4300		3000	1200	4730	5200	1.05	1.04	20.81	29.06	na	na
	WD-6000		3500	1500	5500	7500	1.41	1.36	22.78	29.22	na	na
WG												
513	WG-1600	5.13	800	800	1385	1800	0.87	0.82	45.47	58.00	X	na
	WG-2500		1800	1800	2565	3100	1.23	1.22	41.42	53.95	X	na
	WG-4000		2800	1200	4560	5200	1.67	1.66	34.58	57.43	na	na
	WG-5600		4000	4000	6460	7200	1.84	1.83	26.71	45.84	na	na
	WG-10000		7000	7000	10350	12000	3.69	3.64	33.17	41.75	na	na
WE												
538	WE-1500	5.38	800	200	1720	2300	1.38	1.22	54.41	73.45	na	X
	WE-2600		1500	600	2544	3200	1.43	1.31	47.31	63.20	X	na
	WE-3600		2400	2500	3489	4600	1.91	1.85	50.55	64.29	X	na
	WE-5000		2500	1500	5000	6000					na	X
	WE-5500		3500	1500	5799	7300	2.82	2.74	45.99	61.93	na	X
	WE-7000		4000	2000	7388	10000	3.44	3.36	45.21	63.51	na	X
	WE-8500		6414	2500	8850	10565	3.43	3.30	36.64	62.57	na	X
	WE-11000		8000	3000	11340	14000	4.43	4.36	38.06	50.05	na	X
WH												
562	WH-15000	5.62	10000	5000	15500	18000	8.67	8.60	54.06	72.12	na	X
	WH-20000		12000	6000	19662	25500	11.80	10.89	52.99	72.87	na	X
	WH-30000		21000	9000	29080	36000	15.20	14.80	48.23	53.62	na	X
WJ												
675	WJ-9000	6.75	6000	6000	9560	13000	8.94	7.80	86.27	116.54	na	X
	WJ-12000		7500	7500	12544	17000	11.61	9.79	84.50	103.49	na	X
	WJ-16000		12000	12000	14500	19000	10.35	9.96	69.80	102.60	na	X

Known Variables:
 Casing – 7", 29lbs/ft
 Desired Rate – 2000bpd
 Calculated TDH : 5,500 ft

538	WE-1500	5.38	800	200	1720	2300
	WE-2600		1500	600	2544	3200
	WE-3600		2400	2500	3489	4600
	WE-5000		2500	1500	5000	6000
	WE-5500		3500	1500	5799	7300
	WE-7000		4000	2000	7388	10000
	WE-8500		6414	2500	8850	10565
	WE-11000		8000	3000	11340	14000

1st Option: 538 Series WE-1500

Head/Stage = 51.5 Ft
HP/Stage = 1.28
Efficiency = 58 %



Step 5. Select pump type

Determine Total Stages BHP/Efficiency

$$\#stages = \frac{TDH}{Head / Stg}$$

$$\#stages = \frac{5,500}{51.5} = \mathbf{107}$$

$$TotalBHP = \#stages \times HP / Stg$$

$$TotalBHP = 114 \times 1.28 = \mathbf{137 \text{ BHP}}$$

Known Variables:

WE1500 head per stage – 51.5ft

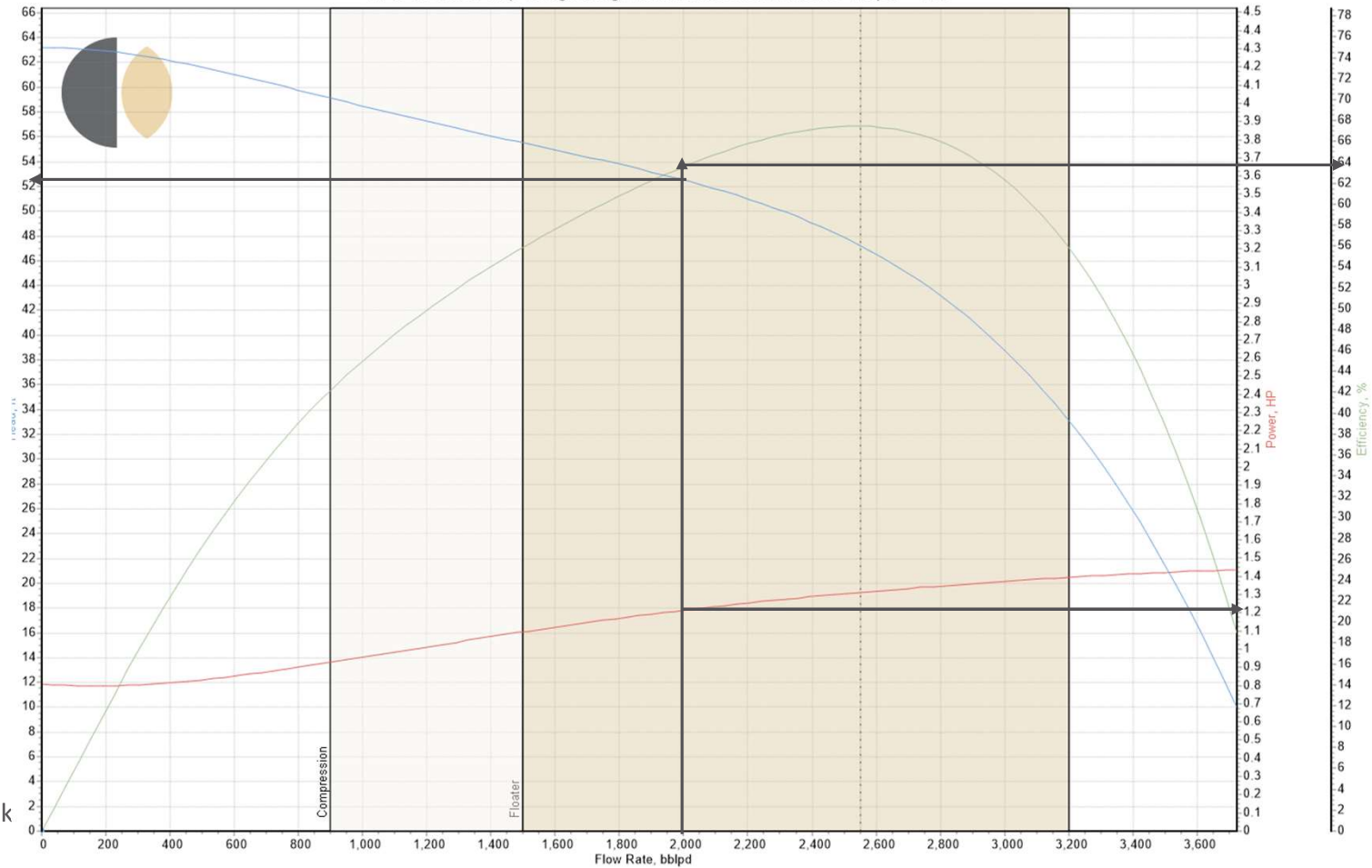
WE1500 BHP per stage – 1.28



2nd Option 538 Series WE-2600

Head/Stage = 52.5Ft
HP/Stage = 1.20
Efficiency = 64 %

APC WE-2600 Pump 1 Stage Design Performance at 60 Hz 3500 RPM SpGr = 1.00



Step 5. Select pump type

Determine Total Stages BHP/Efficiency

$$\#stages = \frac{TDH}{Head / Stg}$$

$$\#stages = \frac{5,500}{52.5} = \mathbf{105 \text{ stg}}$$

$$TotalBHP = \#stages \times HP / Stg$$

$$TotalBHP = 105 \times 1.20 = \mathbf{126 \text{ BHP}}$$

Known Variables:

WE2600 head per stage – 52.5ft

WE2600 BHP per stage – 1.20

Step 5. Check housing capacity & Thrust

APC 538 Series WE-2600 Pumps

APC 538 Series WE-1500 Pumps

Engineering data

Rate	1500
Stage Type	Mixed flow
Housing Diameter, inch	5.38
Shaft Diameter, inch	0.875
Min Casing Size, inch	7
Shut-in Comp. Thrust, lb	45
Burst Pressure STD, psi	5000
Burst Pressure High, psi	6000
Integral Intake	

Performance data

Min ROR CMP, bblpd	560
Min ROR FLT, bblpd	800
BEP Rate, bblpd	1903
Max ROR, bblpd	2300
Max Efficiency, %	58.444
Mah Head, ft	69.54
Max Graph Rate, bblpd	3120

Engineering data

Rate	2600
Stage Type	Radial flow
Housing Diameter, inch	5.38
Shaft Diameter, inch	0.875
Min Casing Size, inch	7
Shut-in Comp. Thrust, lb	96.3
Burst Pressure STD, psi	5000
Burst Pressure High, psi	6000
Integral Intake	

Performance data

Min ROR CMP, bblpd	900
Min ROR FLT, bblpd	1500
BEP Rate, bblpd	2550
Max ROR, bblpd	3200
Max Efficiency, %	67.475
Mah Head, ft	63.2
Max Graph Rate, bblpd	3750

Pump Design	AR COMPRESSION		
Housing	Length, ft	Weight, lb	Stages
10	2.1	117	4
20	3.5	178	11
30	4.9	245	18
40	6.3	312	26
50	7.7	373	33
60	9.1	440	40
70	10.5	502	47
80	11.9	569	54
90	13.3	635	61
100	14.7	697	68
110	16.1	764	76
120	17.5	825	83
130	18.9	892	90
140	20.3	959	97
150	21.8	1021	104

Total Thrust = Thrust/Stage * No. of STG's

WE1500 Option = 45 * 108 = 4,860 lb.

WE2600 Option = 96.3 * 105 = 10,207 lb.

Pump Design	AR COMPRESSION		
Housing	Length, ft	Weight, lb	Stages
10	2.1	77	7
20	3.5	145	19
30	4.9	213	31
40	6.3	272	42
50	7.7	340	53
60	9.1	407	65
70	10.5	467	76
80	11.9	535	88

- WE1500 AR-CP will be required 107 stages
- Either 2 HSG (54*2)= 108STG or other combinations

- WE2600 AR-CP will be required 105 stages
- Either 2 HSG (53*2)= 106 STG or other combinations

Step 5. Select pump type

Decision point factors

- Horsepower.
- Pump thrust.
- Pump Efficiency.
- Shaft HP limitations.
- Design Target.
- Well problems.
- Cost \$
- Expected future scenario.
- Field Experience & History.
- Customer preference
- Availability.



Sequence

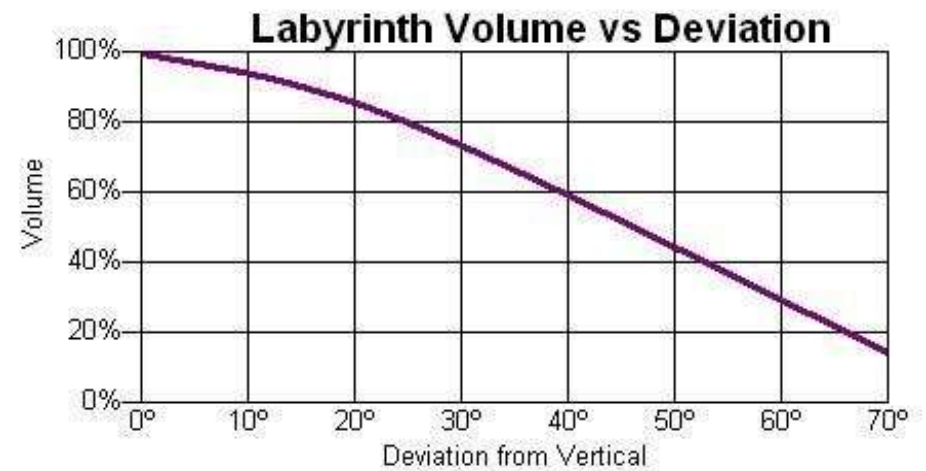
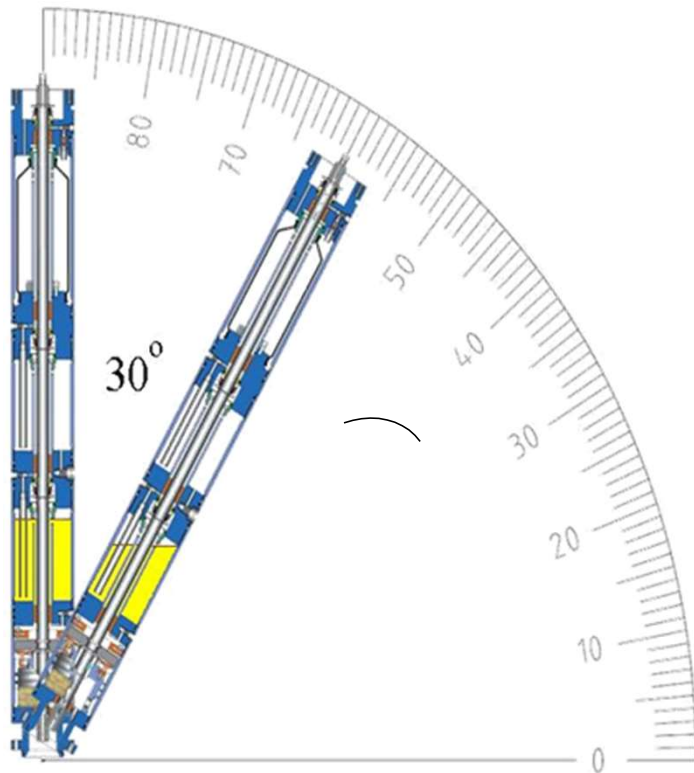
- Step 1. • Collect and Analyze Data
- Step 2. • Determine Pump Intake Pressure
- Step 3. • Gas Calculations
- Step 4. • Determine Total Dynamic Head
- Step 5. • Select Pump Type
- **Step 6. • Select Seal and Motor**
- Step 7. • Select Power Cable
- Step 8. • Select Motor Controller

Step 6: Seal selection

- Seal requirements
 - Sufficient volume capacity
 - Thrust bearing capable to handle pump thrust
 - Bag sections for deviated well
 - Proper O-rings and seals for fluid and temperature
 - Proper combinations.



Step 6: Seal selection



Step 6. Select seal

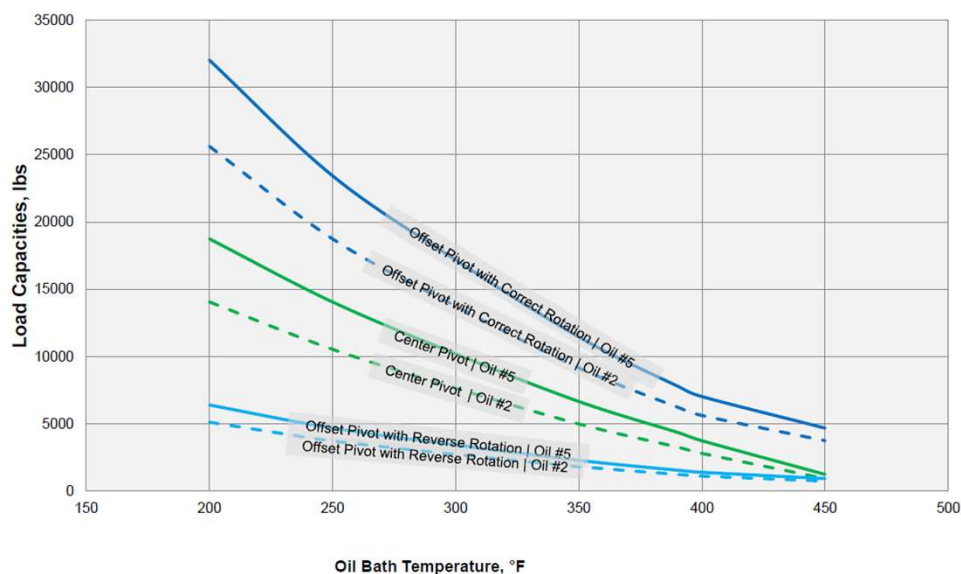
Seal selection criteria

- Motor series
- Options
 - Single vs double seals
 - Bag vs labyrinth
 - Well deviation
 - Oil temp rating
 - BHT
 - Thrust load
 - Corrosion resistance
 - Adapters
 - specific gravity of fluids



Step 6. Pro seal Specifications

540 Series ProSeal TB Load Capacities | SF=2 | 60 Hz



APC 540 Series Proseals

Engineering data

Housing Diameter, inch	5.125
Shaft Diameter, inch	1.188
Min Casing Size, inch	7
Max Thrust STD @60Hz/250F	2459
Max Thrust HL @60Hz/250F	14063

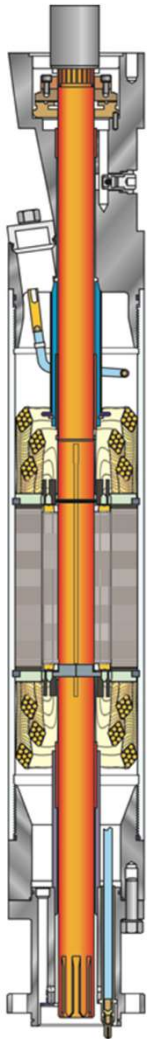
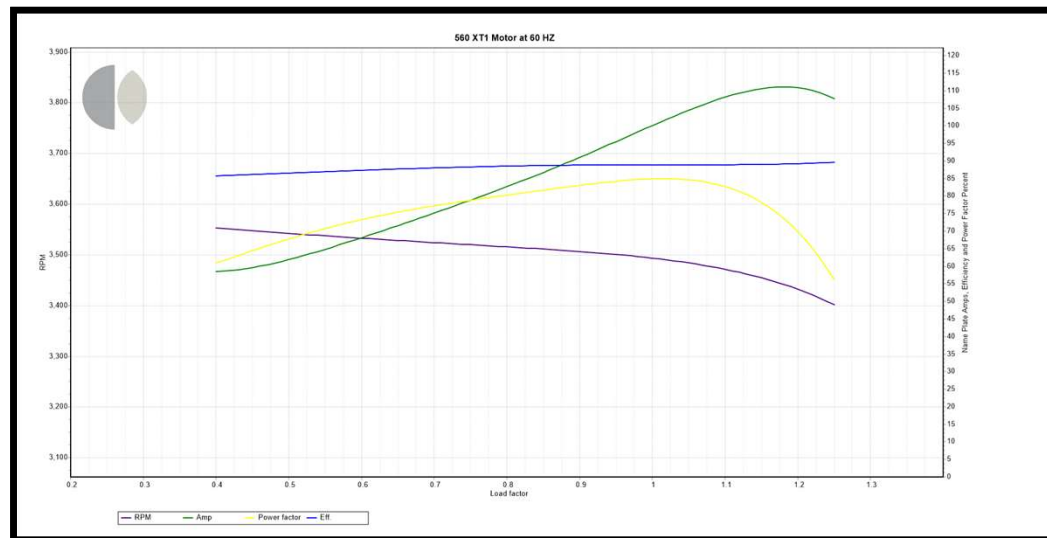
Description

540	PS-B-HL AR	Single Bag Type - HL AR
540	PS-BsL-HL AR	Bag - Labyrinth Type - HL AR
540	PS-BsB-HL AR	Double Bag Type - HL AR
540	PS-BpB-HL AR	Double Bag Type - HL AR
540	PS-BsBsL-HL AR	Double Bag - Labyrinth Type - HL AR
540	PS-BpBsL-HL AR	Double Bag - Labyrinth Type - HL AR
540	PS-LsBpB-HL AR	Labyrinth - Double Bag Type - HL AR
540	PS-LsBsB-HL AR	Labyrinth - Double Bag Type - HL AR
540	PS-BpBsLsL-HL AR	Double Bag-Double Labyrinth Type-HL AR
540	PS-BsLsBsL-HL AR	Double Bag-Double Labyrinth Type-HL AR
540	PS-BsBsLsL-HL AR	Double Bag-Double Labyrinth Type-HL AR

Step 6. Motor Selection

- Motor Considerations

- Horsepower
- Voltage
- Amperage
- Load
- Bottomhole temp. – BHT
- Cost \$
- Availability.





Step 6. Motor selection

Determine total HP requirement

$$\text{Total HP} = \text{BHP} + \text{intake/GS HP} + \text{Seal HP}$$

$$\text{Total HP} = 137 * 1.05 + 5 \text{ hp} = 149 \text{ hp}$$

Known Variables:

- BHP 137
- SGL 1.05
- Intake HP – 1.0
- Seal BHP - 4

Step 6: Motor selection

- Selected Motor will be

HP (Δ)	Volt Δ	Amp Δ
157	1216	77.7
157	2433	38.9
157	3043	30.7
183	1183	92.2
183	2376	47
183	3549	30.7
209	1073	116.1
209	1350	92.2
209	2156	57.8
209	2433	51.5
209	4054	30.7

Known Variable:

Casing Size: 7"

Required HP = 149

BHT – 180f

➤ Motor Loading at Design Point $\pm 80\%$ of nameplate.

- 157HP motor – Load 95%
- 183HP motor – Load 81%
- 209HP motor – Load 71%

➤ Recommended High Volt/ Low Amp as 1st Option.

- 183HP XT1 selected
 - Volt/Amps: 2376/47
 - Length: 17.5ft
 - Temperatures up to 325 F
 - Motor load: 81%

Sequence

- Step 1. • Collect and Analyze Data
- Step 2. • Determine Pump Intake Pressure
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- Step 4. • Determine Total Dynamic Head
- Step 5. • Select Pump Type
- Step 6. • Select Seal and Motor
- **Step 7. • Select Power Cable**
- Step 8. • Select Motor Controller



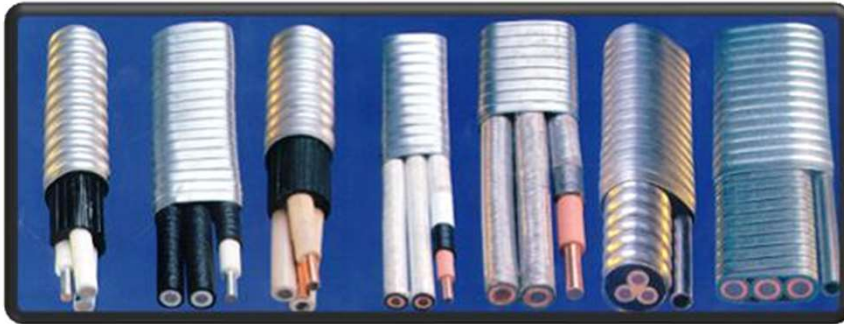
Cable Selection Criteria

- Amperage (voltage drop)
- Conductor temperature (insulation material)
- Insulation voltage rating
- Gas handling (decompression protection)
- Corrosive properties of well fluid
- Available space (casing clearance)
- Inventory availability and standard offerings



Cable Selection Criteria

- **Amperage (voltage drop)**
- Conductor temperature (insulation material)
- Insulation voltage rating
- Gas handling (decompression protection)
- Corrosive properties of well fluid
- Available space (casing clearance)
- Inventory availability and standard offerings





Step 7. Select power cable

Amperage (voltage drop)

Line loss is the normal reduction in available voltage after long distance transmission

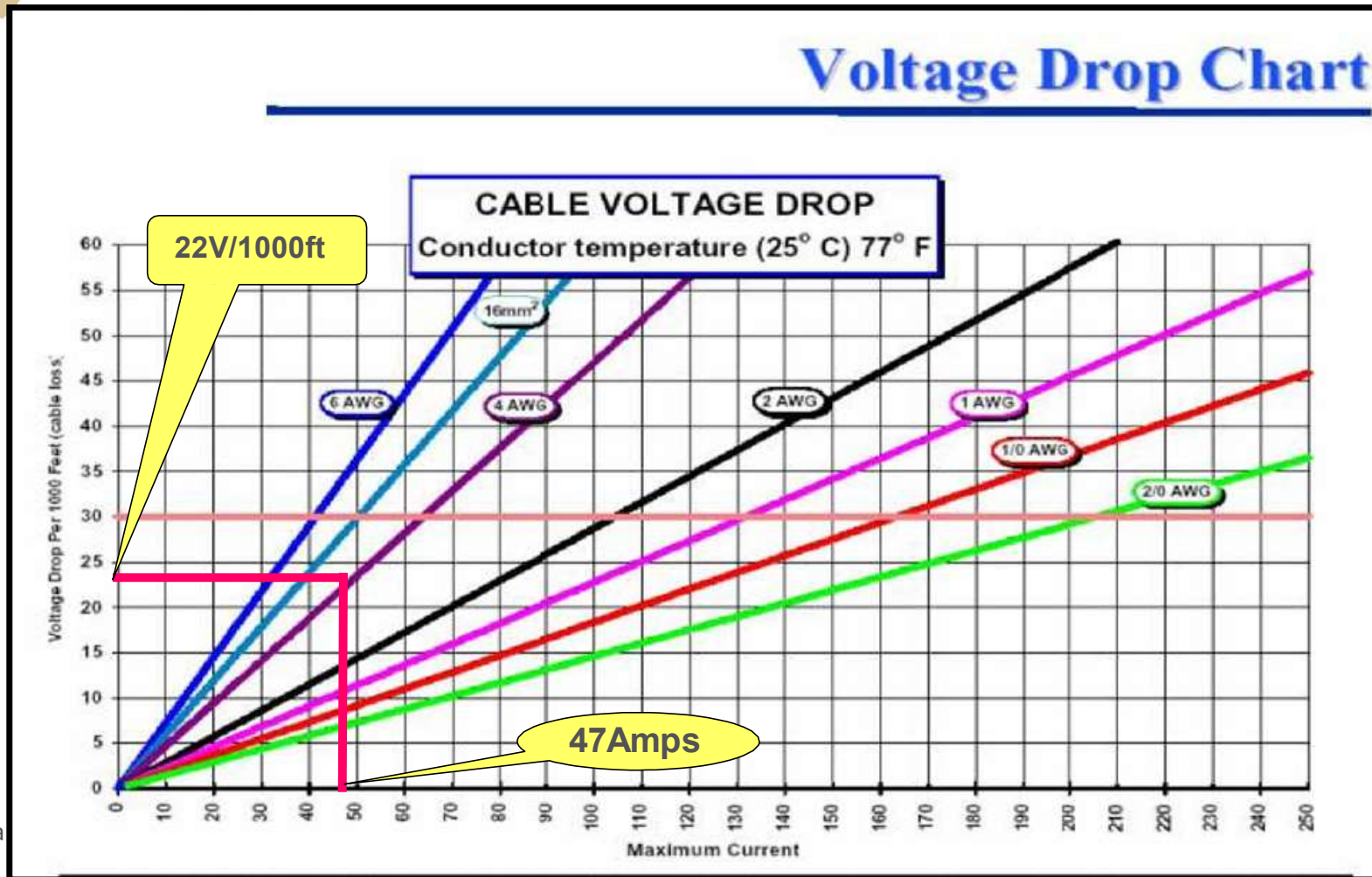
High temperatures impede the ability of the conductor to transmit voltage (resistance)

Voltage drop refers to voltage losses due to distance and temperatures



Step 7. Select Power Cable

Known Variables:
Motor Amps – 47
Nameplate Volts - 2376



Step 7. Select power cable

Temperature Factor: 1.23

Known Variables:
Temp – 180°F

CONDUCTOR VOLTAGE LOSS TABLE (Temperature Correction Factor @ 77F)								
Temp °F	Temp °C	Mult. Factor	Temp °F	Temp °C	Mult. Factor	Temp °F	Temp °C	Mult. Factor
50	10	0.94	221	105	1.31	410	210	1.71
59	15	0.96	230	110	1.33	419	215	1.73
68	20	0.98	239	115	1.35	428	220	1.75
77	25	1.00	248	120	1.37	437	225	1.77
86	30	1.02	257	125	1.39	446	230	1.79
95	35	1.04	266	130	1.40	455	235	1.81
104	40	1.06	275	135	1.42	464	240	1.83
113	45	1.08	284	140	1.44	473	245	1.85
122	50	1.10	293	145	1.46	482	250	1.87
131	55	1.12	320	160	1.52	491	255	1.89
140	60	1.13	329	165	1.54	500	260	1.91
149	65	1.15	338	170	1.56	509	265	1.92
158	70	1.17	347	175	1.58	518	270	1.94
167	75	1.19	356	180	1.60	527	275	1.96
176	80	1.21	365	185	1.62	536	280	1.98
185	85	1.23	374	190	1.64	545	285	2.00
194	90	1.25	383	195	1.65	554	290	2.02
203	95	1.27	392	200	1.67	563	295	2.04
212	100	1.29	401	205	1.69	572	300	2.06

Step 7. Select power cable

Amperage (voltage drop)

Known Variables:
Line Loss – 22V/1000ft
Temp. Factor – 1.23
Pump Depth – 5200ft
Surface Cable – 200ft

$$\begin{aligned} \text{VoltageDrop} &= \text{LineLoss}(\text{TempFactor}) \\ &= 22 \times 1.23 \\ &= 27.06 \text{Volts} / 1000 \text{ft} \end{aligned}$$

$$\begin{aligned} \text{TotalVoltageDrop} &= (\text{VoltageDrop})((\text{Depth} + \text{Surf.Cable}/1000)) \\ &= (27.06 \text{Volts})((5200 + 200\text{ft})/1000) \\ &= 146 \text{Volts} \end{aligned}$$

Step 7. Select power cable

Selection analysis

Voltage Drop < 30 volts per 1000 ft ☒

Voltage Drop = 27 volts / 1000 ft

Voltage Drop < 15% of nameplate voltage ☒

146 volts / 2376 volts = 6.14%

Known Variables:

Nameplate Volts – 2376

Voltage Drop – 146 Volts

Motor Depth – 5200 ft

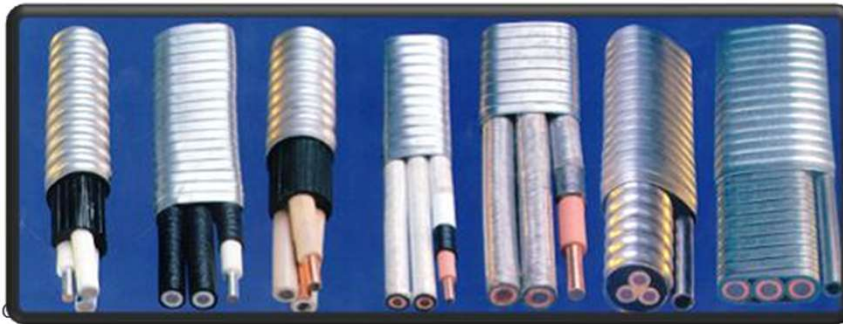
Surface Cable – 200 ft

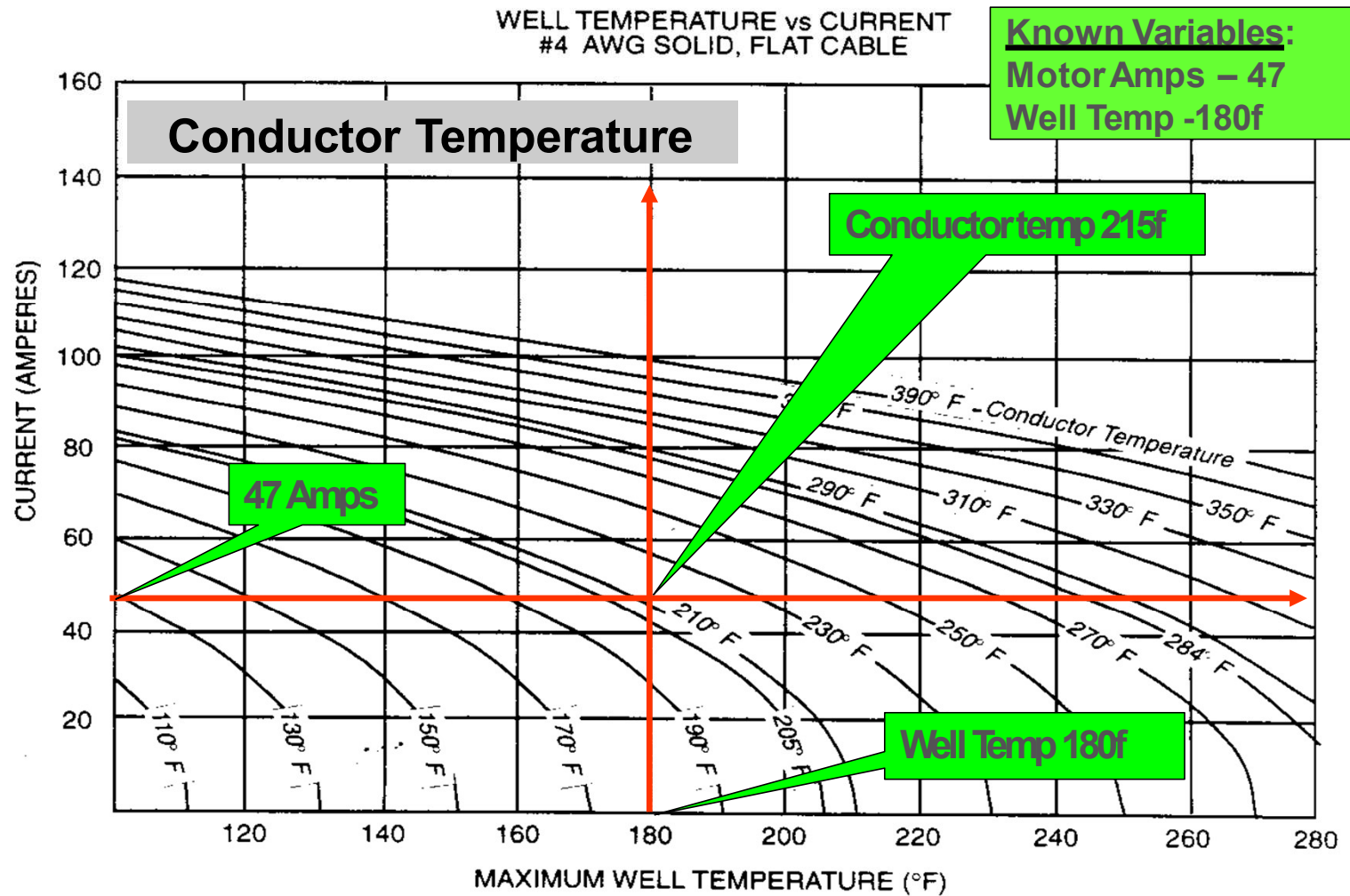
Cable Gauge - #4

#4 AWG conductor is used in this application

Cable Selection Criteria

- Amperage (voltage drop)
- **Conductor temperature (insulation material)**
- Insulation voltage rating
- Gas handling (decompression protection)
- Corrosive properties of well fluid
- Available space (casing clearance)
- Inventory availability and standard offerings





Conductor temperature

Temperature Ratings of ESP Cables - insulation / covering / jacket (*denotes limiting material)

• Polypropylene* / Nitrile	205° F	(96° C)
• Polypropylene* / Lead	257° F	(125° C)
• EPDM / Nitrile* (with or w/o Barrier)	280° F	(140° C)
• EPDM / Low Temp Barrier* / EPDM	300° F	(150° C)
• EPDM / EPDM (with or w/o hi temp Barrier)	400° F	(205° C)
• EPDM* / Lead Sheath	450° F	(230° C)
• Motor Lead Extension - Low Temp	250° F	(120° C)
• Motor Lead Extension - High Temp	400° F	(205° C)
• Motor Lead Extension - Lead Sheath	450° F	(230° C)

EPDM insulation is used in this application as the conductor temperature is beyond the upper limit for polypropylene



Cable Selection Criteria

Insulation Type:

Polypropylene (PPE), also known as polypropene, is a thermoplastic polymer, made by the chemical industry and used in a wide variety of applications

PPE, considered a thermoplastic material, is characterized as follows:

- Low temperature rating (250 degf)
- Excellent resistance to well fluid
- Excellent electrical properties
- Cost effective
- Is crazing (microcracking) when exposed to high levels of CO₂ (>5%)

EPDM, an elastomeric rubber commonly used as the primary insulation on downhole electrical cable.

EPDM, a thermoset, elastomeric material, is characterized as follows:

- High temperature rating (450 degf)
- Excellent electrical properties, though perhaps not as good as PPE
- Limited resistance to well fluids (oil swell).

EPDM compounds are chosen because they have extremely good electrical properties and temperature capabilities.

Cable Selection Criteria

- Amperage (voltage drop)
- Conductor temperature
- **Insulation voltage rating**
- Gas handling (decompression protection)
- Corrosive properties of well fluid
- Available space (casing clearance)
- Inventory availability and standard offerings



Step 7. Select power cable

Surface voltage requirement

Calculate surface voltage

SurfaceVolts = motor volts + voltage drop

$$\text{SurfaceVolts} = 2376 + 146 = 2522$$

Known Variables:

Nameplate Volts – 2376

Voltage Drop – 146 Volts

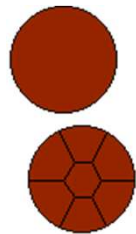
Insulation KV options:

- 3 kv
- 4 kv
- 5 kv

3 or 5 kv insulation is suited to the application

Cable Selection Criteria

Conductor Configuration:



Configuration	Advantage	Disadvantage
Solid	• Smallest diameter • Lowest cable cost • Low electrical stress	• Decreased flexibility , especially with larger conductors
Compact-strand	• Larger diameter than solid but smaller than strand • Good flexibility	• Increased cable cost due to more material usage and additional processing steps

Cable Selection Criteria

- Amperage (voltage drop)
- Conductor temperature
- Insulation voltage rating
- **Gas handling (decompression protection)**
- Corrosive properties of well fluid
- Available space (casing clearance)
- Inventory availability and standard offerings



Step 7. Select power cable

Gas Handling (decompression)

Extruded Fluorobarrier

- Slows movement of gas into and out of insulation

Lead Sheath

- Prevents entry of gas into insulation
- Suitable for use in H₂S environments



Cable Selection Criteria

- Amperage (voltage drop)
- Conductor temperature
- Insulation voltage rating
- Gas handling (decompression protection)
- **Corrosive properties of well fluid**
- Available space (casing clearance)
- Inventory availability and standard offerings



Step 7. Select power cable

Aarmor Selection:

Galvanized Steel (MOST COMMON)

- Can upgrade to heavy or double galvanized steel (Mechanical Protection)

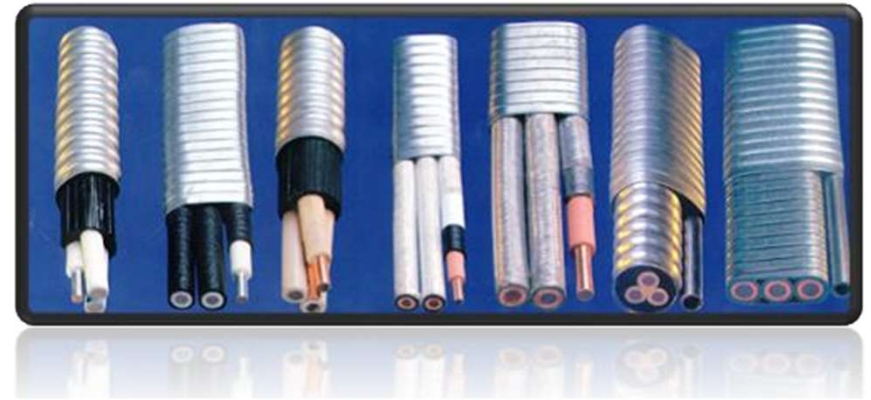
Stainless Steel (special applications)

- Avoid when well temp is above 250°F or the steel may suffer from corrosion and become brittle.
- In the presence of chlorides at temperatures above 160°F stress corrosion cracking will occur

Monel (Corrosive Situations)

- Low pH wells, high CO₂ or H₂S = ONLY CHOICE
- Standard for MLEs because of the higher temps and potential for corrosion in the bottom of the well fluid
- Recommended Minimum Monel thickness is 0.02" as lower than this thickness, the interlock will break when the cable is reused.
- The weight may damage the armor of the cable on the bottom layer on the reel

- Amperage (voltage drop)
- Conductor temperature
- Insulation voltage rating
- Gas handling (decompression protection)
- Corrosive properties of well fluid
- **Available space (casing clearance)**
- Inventory availability and standard offerings



Step 7. Select power cable

Known Variables:
Casing – 7" 29#
Tubing – 2⁷/₈

Clearance:

Clearance = (Casing drift ID – Tubing coupling OD)

$$\text{Clearance} = (6.148 - 3.668) = 2.48$$

Profile Options:

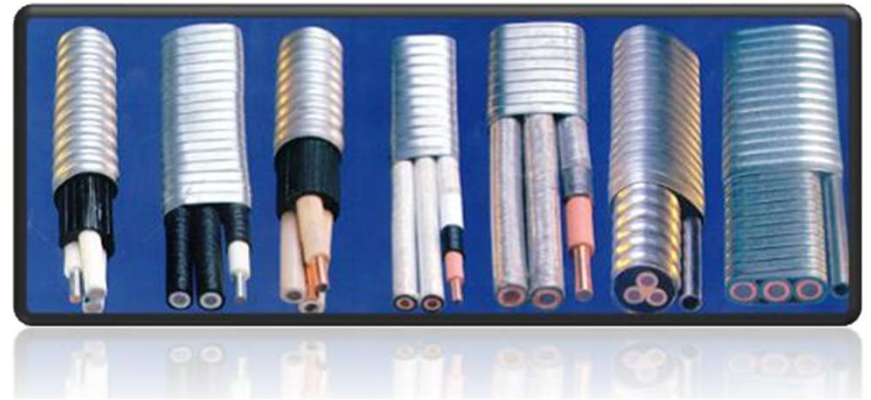
- Flat: Approximately 0.7"
- Round: Approximately 1.19"

Flat or round options are both suited



Cable Selection Criteria

- Amperage (voltage drop)
- Conductor temperature
- Insulation voltage rating
- Gas handling (decompression protection)
- Corrosive properties of well fluid
- Available space (casing clearance)
- **Inventory availability and standard offerings**



- Step 1. • Collect and Analyze Data
- Step 2. • Determine Pump Intake Pressure
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- Step 7. • Select Power Cable
- **Step 8.** • **Select Motor Controller**



Step 8. Select motor controller

Determine Surface Voltage

Known Variables:

Nameplate Volts – 2376
Voltage Drop – 146 Volts
Motor Amps – 47 Amps

Surface Voltage = Nameplate Voltage + Voltage Drop

Surface Voltage = 2,376 Volts + 146 Volts = **2,522 Volts**



Step 8. Select motor controller

Determine Kilovolt Amps- **Surface KVA**

Known Variables:
Motor Amps – 47 Amps
Surface Volts – 2522 V

$$KVA = \frac{(Surface Voltage)(Motor Amps)(1.73)}{1000}$$

$$KVA = \frac{(2522 Volts)(47 Amps)(1.73)}{1000} = 205 KVA$$

- ✓ Select Transformer with minimum 250 KVA, 3 φ, and multi tapped secondary

Step 8. Select motor controller

Determine Drive Amperage Requirement

Known Variables:
Nameplate Volts – 2376
Voltage Drop – 146
Surface Voltage – 2522
Motor Amps – 47

- Determine step-up ratio:
 - Divide surface volts by drive output voltage
 - $2522 / 480 = 5.25$
 - Multiply step-up ratio by motor amperage
 - $5.25 \times 47 = 247$ amps
- ✓ Select Transformer with minimum 250 KVA, 3 ϕ , and multi tapped secondary

Transformers

Used to change voltage from one value to another

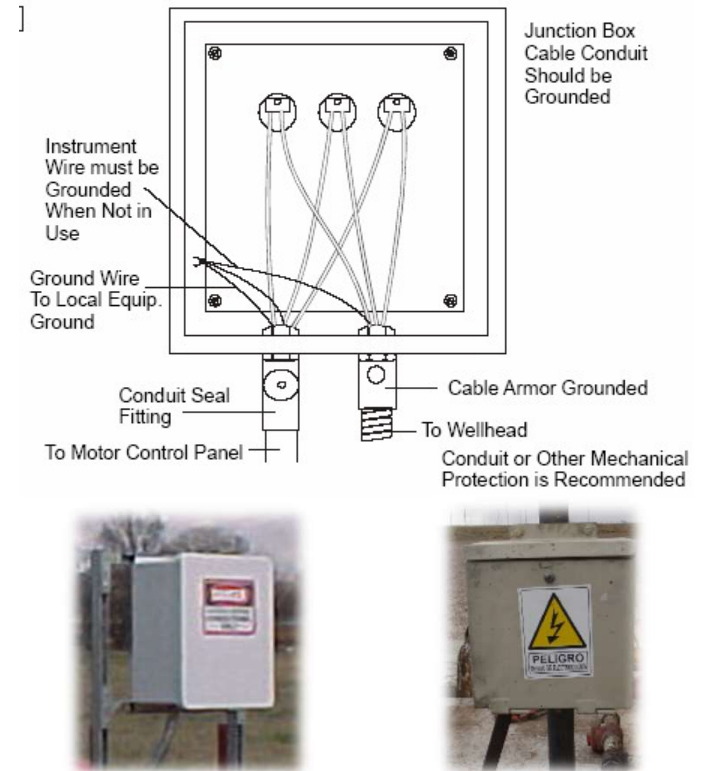
Step-up style takes incoming (primary) voltage and increases it for output

Step-down style lowers primary voltage



Junction box

- Provides a convenient connection point for the downhole power cable and the surface electrical cable
- Vents downhole gasses that escape through the cable insulation and jacket
- Provides a test point by splitting the downhole and surface equipment



The background is a dark, moody photograph of an industrial facility at sunset. Silhouettes of buildings and a tall lattice tower are visible against a sky filled with clouds. A bright orange and yellow sunset is visible through a heart-shaped graphic in the center. The heart is composed of two overlapping semi-circles, each outlined in white. The text "THANK YOU" is written in white, bold, sans-serif capital letters across the middle of the heart.

THANK YOU